NSW Road Crash Severity Prediction: Project Plan and EDA

Group 7, Workshop 12

Introduction

```r
library(ggplot2)
library(dplyr)
library(readxl)
library(forcats)
library(scales)
library(knitr)
```

```r
df_merged <- read.csv("nsw_road_crash_merged_2020_2024.csv", check.names = FALSE)
```

1 Framing & Dataset Selection

1.1 Classification Task

Based on environmental conditions and traffic unit characteristics, predict the severity of a road crash as a multi-class classification problem: Fatal, Injury, or Non-casualty.

1.2 Research Question

Given environmental conditions (weather, natural lighting, road surface, speed limit) and traffic unit characteristics (number of traffic units involved, type of traffic units, road classification), can we predict the severity of a road crash in NSW?

1.3 Motivation

This research question has direct real-world relevance, NSW recorded 340 road fatalities in 2024, with over two-thirds occurring in regional areas (Transport for NSW, 2025), highlighting the need for data-driven approaches to understanding crash severity. Our study affects the following groups:

NSW Traffic Management Authorities: By identifying key combinations of environmental and traffic characteristics that contribute to severe crashes, authorities can make targeted decisions such as adjusting speed limits, improving road design, and increasing enforcement during high-risk conditions.

Emergency Response Teams: In the early stages of a crash, predicting severity based on environmental conditions can help allocate ambulance and emergency resources more quickly and effectively.

Policymakers and Insurance Institutions: Understanding which traffic unit types (e.g. heavy vehicles) pose higher risks under specific environmental conditions can inform more precise road safety policies and insurance pricing models.

The dataset contains 92,189 crash records across NSW from 2020 to 2024, covering variables such as weather, natural lighting, road surface condition, speed limit, number of traffic units involved, and traffic unit types: fully supporting the environmental and traffic unit characteristics required by our research question.

A predictive model for crash severity enables data-driven prioritization, this can help authorities allocate resources design intervention strategies, and form policy based on evidence rather than intuition.

2 Consider & Gather Data

2.1 Data Source

The data used in this study is sourced from the Australian Government open data platform

NSW Crash Data Dataset: https://www.data.gov.au/data/dataset/nsw-2-nsw-crash-data

This dataset is published by Transport for NSW and provides detailed information on road crashes in NSW, including environmental conditions, road characteristics, vehicle types, and injury outcomes (Transport for NSW, 2024).

For this study, we utilized the Crash Dataset (NSW Road Crash Data - 2020-2024 - CRASH.xlsx) and Traffic Unit Dataset (NSW Road Crash Data - 2020-2024 - TRAFFIC UNIT.xlsx) for 2020-2024.The crash dataset contains environmental and road- related variables (e.g. weather, road surface, and speed limit)

- The crash dataset contains:
  - Crash Details (No. Killed, No. Seriously Injured, Severity of Crash)
  - time-related features (Day, Month, etc)
  - location attributes (Direction, latitude, longitude, Distance, etc)
  - environmental conditions (Weather, Natural Lighting, Speed Limit, etc.)
- The traffic unit dataset contains vehicle-level variables:
  - TU type group (states vehicle type)
  - Manoeuvre (states vehicle movement at time of crash)
  - TU role in first impact (states role of vehicle e.g. key traffic unit)
  - TU controlled flag (Whether classified as controlled or uncontrolled)

```r
cat("Merged:", nrow(df_merged), "rows,", ncol(df_merged), "cols\n")
```

```
## Merged: 170962 rows, 59 cols
```

```r
colSums(is.na(df_merged))
```

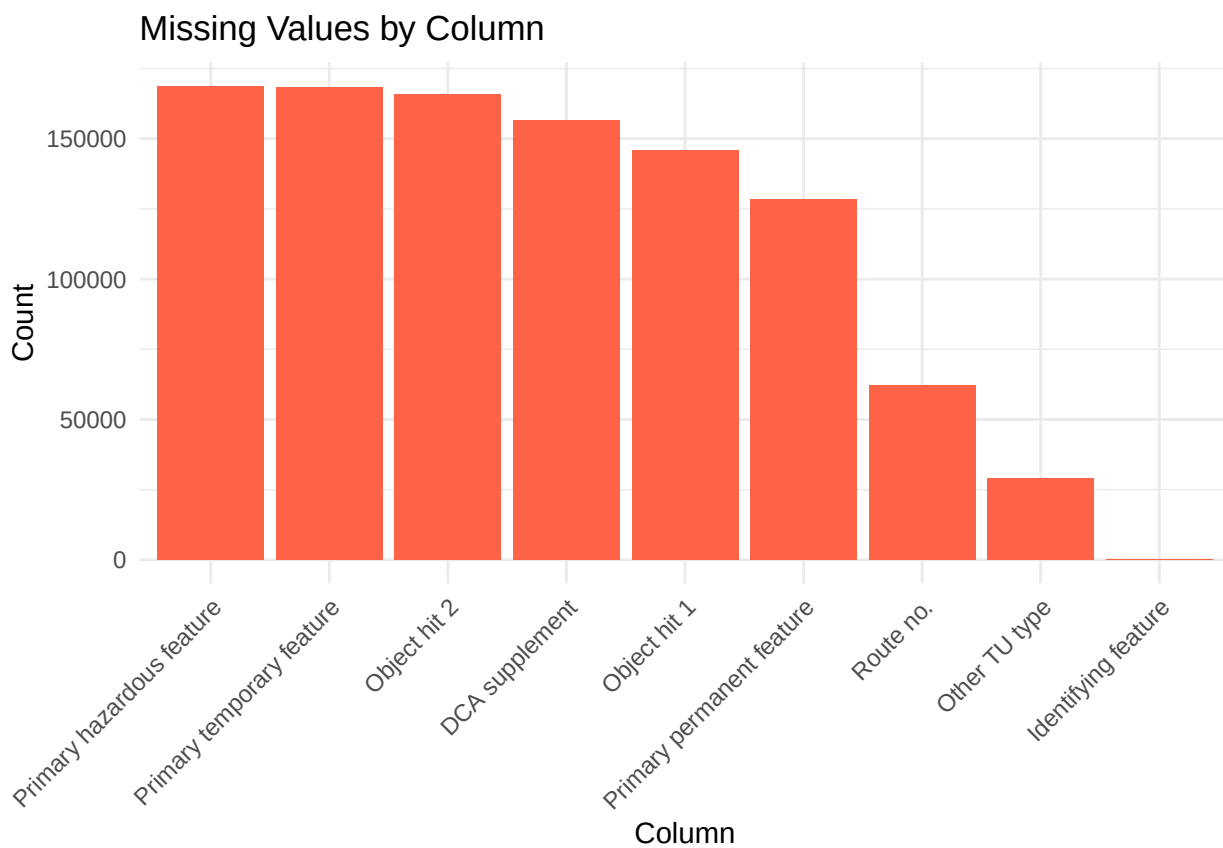
```
##                      Crash ID               Degree of crash
##                             0                             0
##    Degree of crash - detailed                Reporting year
##                             0                             0
##                 Year of crash                Month of crash
##                             0                             0
##          Day of week of crash            Two-hour intervals
##                             0                             0
##               Street of crash                   Street type
##                             0                             0
##                      Distance                     Direction
```

```
##                              0                              0
##           Identifying feature      Identifying feature type
##                              3                              0
##                           Town                     Route no.
##                              0                          62005
##          School zone location            School zone active
##                              0                              0
##              Type of location                       Latitude
##                              0                              0
##                     Longitude                            LGA
##                              0                              0
##                  Urbanisation                  Conurbation 1
##                              0                              0
##                     Alignment     Primary permanent feature
##                              0                         128408
##     Primary temporary feature     Primary hazardous feature
##                         168406                         168803
##               Street lighting                   Road surface
##                              0                              0
##             Surface condition                        Weather
##                              0                              0
##              Natural lighting             Signals operation
##                              0                              0
##         Other traffic control                    Speed limit
##                              0                              0
##   Road classification (admin)                     RUM - code
##                              0                              0
##             RUM - description                     DCA - code
##                              0                              0
##             DCA - description                DCA supplement
##                              0                         156545
##             First impact type                    Key TU type
##                              0                              0
##                 Other TU type No. of traffic units involved
##                          29195                              0
##                    No. killed         No. seriously injured
##                              0                              0
##        No. moderately injured       No. minor-other injured
##                              0                              0
##               Traffic unit ID            TU controlled flag
##                              0                              0
##       TU role in first impact                 TU type group
##                              0                              0
##              Street of travel           Direction of travel
##                              0                              0
##                     Manoeuvre                   Object hit 1
##                              0                         145702
##                  Object hit 2
##                         165996
```

2.2 Dataset Overview

The crash dataset contains 92,189 crash records, and the traffic dataset contains 170,962 records across NSW from 2020 to 2024. The dataset formed by merging both of these on Crash ID contains 170,962 records. The merged dataset is recorded at a vehicle level. This allows vehicle-specific characteristics to be incorporated into the crash severity increasing the richness of our dataset.

```
na_counts <- colSums(is.na(df_merged))
na_df <- data.frame(
  variable = names(na_counts),
  missing = na_counts
) |> subset(missing > 0)

ggplot(na_df, aes(x = reorder(variable, -missing), y = missing)) +
  geom_bar(stat = "identity", fill = "tomato") +
  labs(title = "Missing Values by Column", x = "Column", y = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



2.3 Dataset Justification

The dataset meets several key requirements outlined in the assignment.

Large Dataset(>10,000 observations)

The merged dataset has 170,962 records, providing sufficient scale for robust statistical analysis and model training. The large size enables reliable estimation of relationships between our target variable (crash severity) and its predictors while also supporting validation techniques such as test-train split.

Integrated Dataset (Multiple sources combined)

The dataset is formed by merging crash-level and traffic unit datasets using a common Crash ID. This integration combines environmental, spatial, and vehicle- level information. The process improves the richness of features available for analysis and enables more comprehensive modelling of crash severity.

Messy Dataset (Real-world data challenge)

As a real-world administrative dataset, it contains missing values and potential inconsistencies across variables such as weather, lighting, and vehicle characteristics. Additionally, the process of merging the crash dataset (92,189 records) with traffic unit dataset (170,962 records) exposes existing missing values, as not all variables are available at both the crash and vehicle levels. When merged using Crash ID, crash-level attributes are duplicated across multiple vehicle records, while some vehicle-specific fields may not have corresponding values for all observations.

This results in a dataset with increased sparsity and inconsistencies across columns, requiring preprocessing steps such as handling missing data, managing duplicated information, and cleaning categorical variables. As shown in the above figure, columns such as Primary hazardous feature, Primary temporary feature, and object hit 2 have the highest missingness(>95%) while Route no. and Other TU type have moderate missingness

Multi-class Classification Problem

The target variable, crash severity, is a multi-class outcome derived from Degree of crash column in the crash dataset, consisting of categories such as Fatal, Injury, and Non-casualty. These categories are supported by injury-related variables in the crash dataset (e.g., number killed and number seriously injured). This introduces challenges such as class imbalance, which must be considered when evaluating model performance.

2.4 Summary

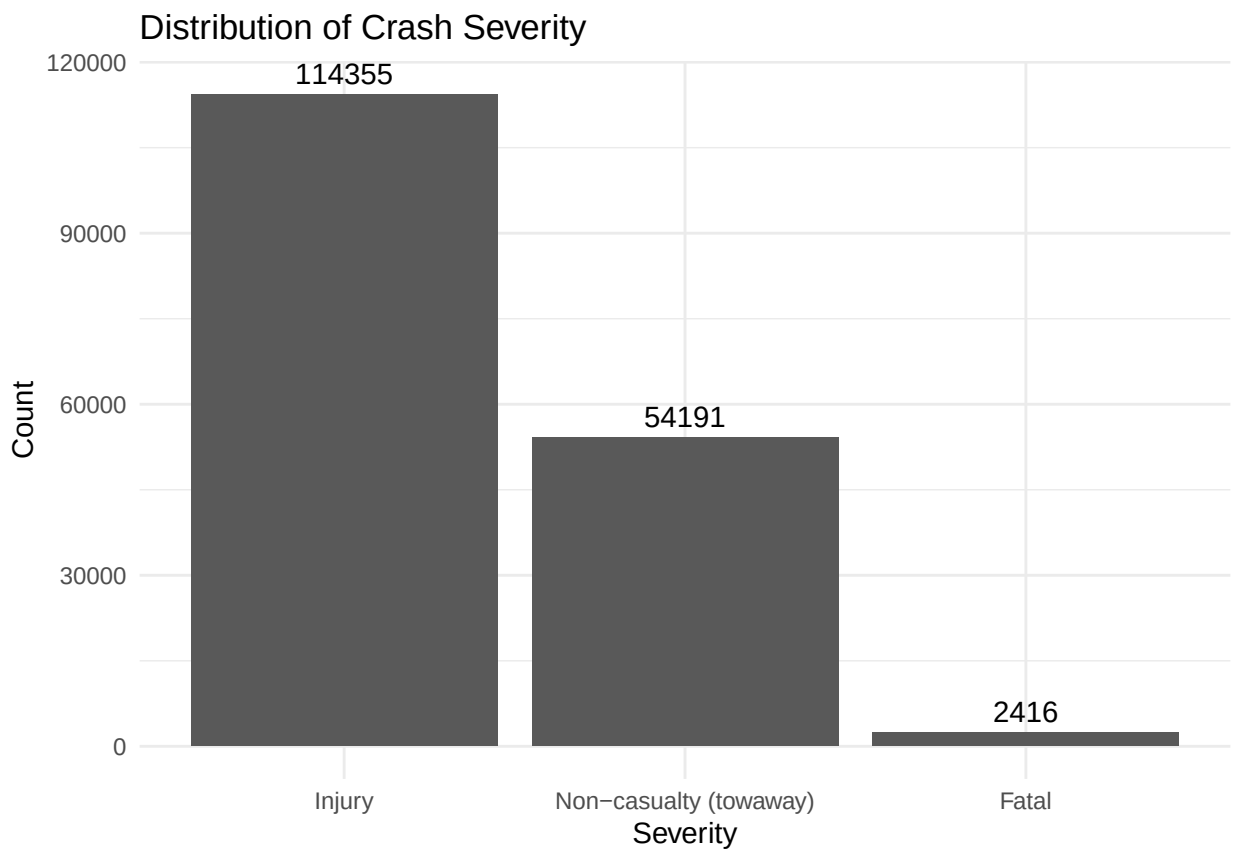
Overall, the dataset is well-suited for this study as it is large, integrated, messy, and supports a multi-classification problem. Its combination of environmental, temporal, spatial, and vehicle-level features provides a strong foundation for modelling and predicting crash severity.

3 Exploratory Data Analysis (EDA)

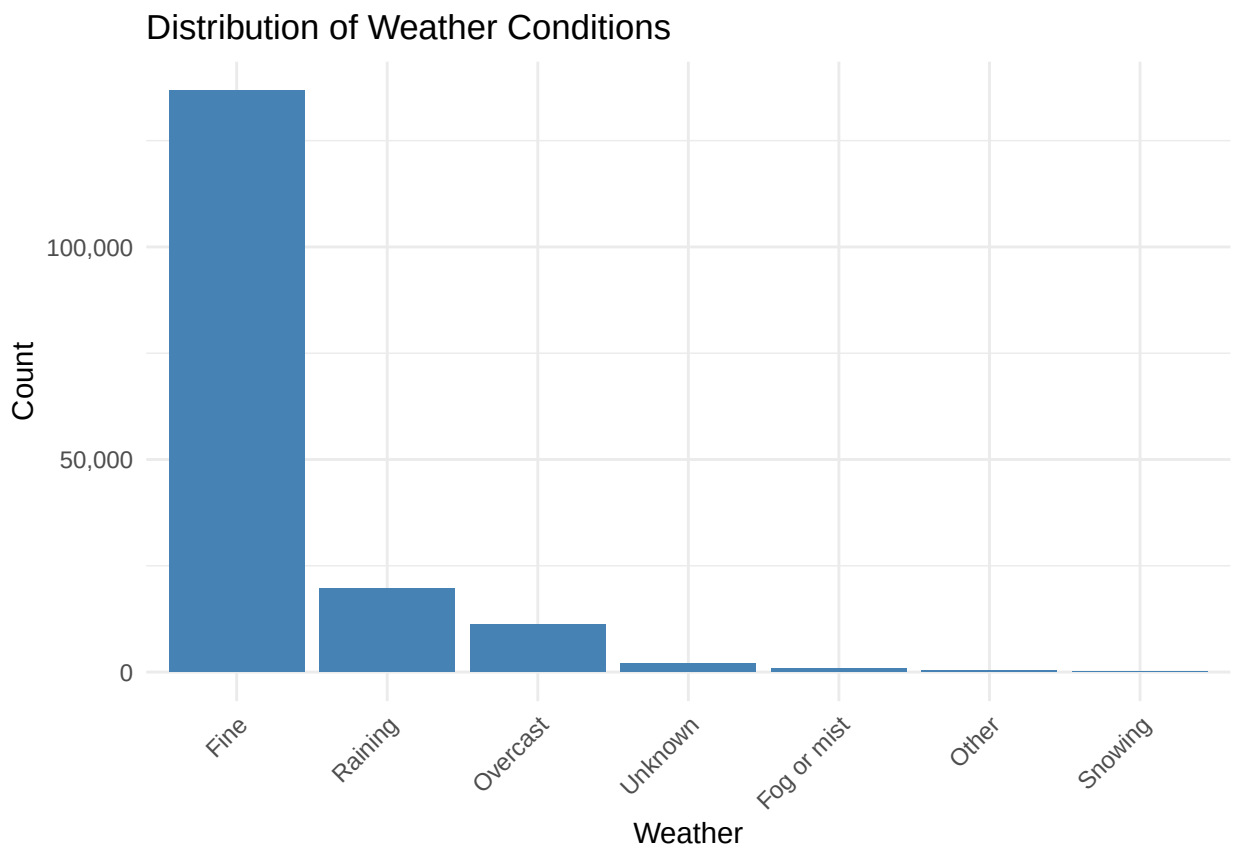
3.1 Univariate Analysis

3.1.1 Distribution

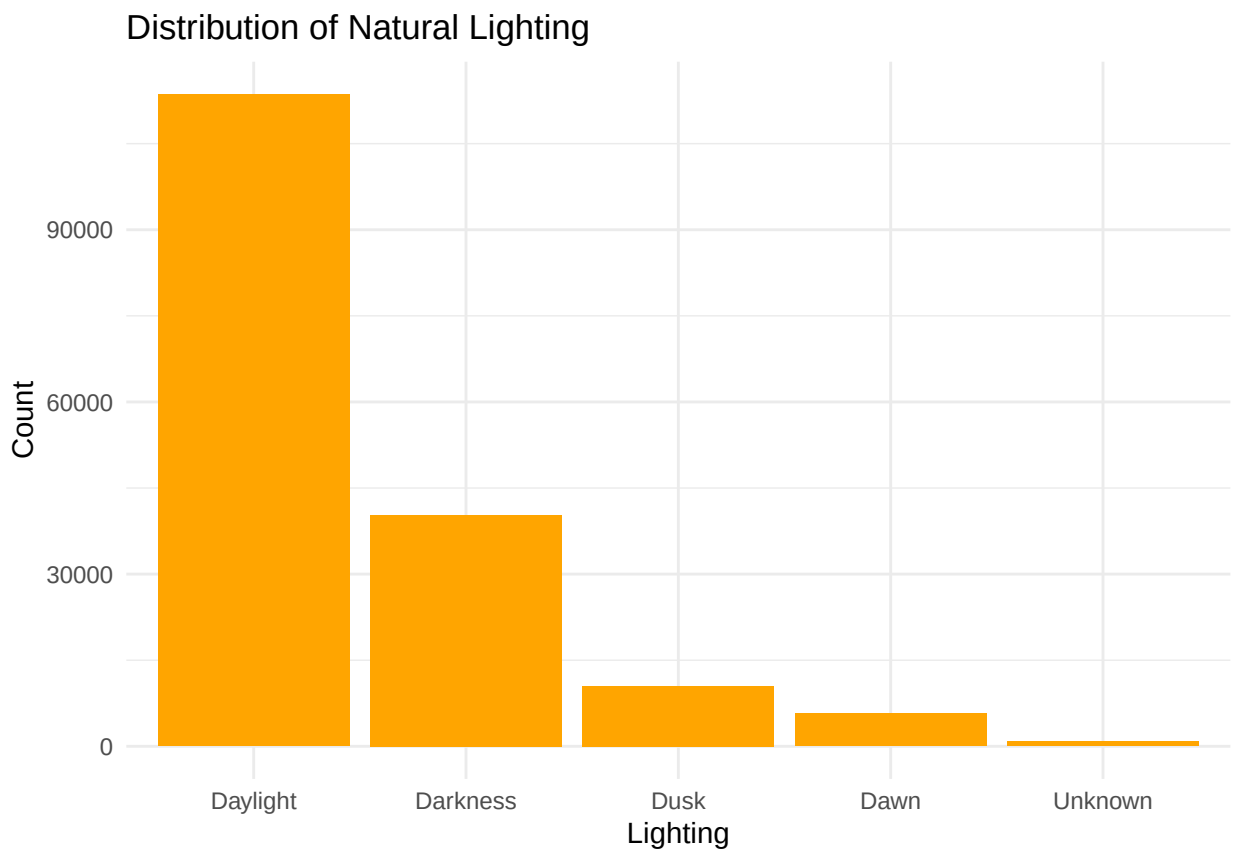
```
ggplot(df_merged, aes(x = fct_infreq(`Degree of crash`))) +
    geom_bar() +
    geom_text(stat = "count", aes(label = after_stat(count)), vjust = -0.5) +
    labs(title = "Distribution of Crash Severity", x = "Severity", y = "Count") +
    theme_minimal() + theme(legend.position = "none")
```



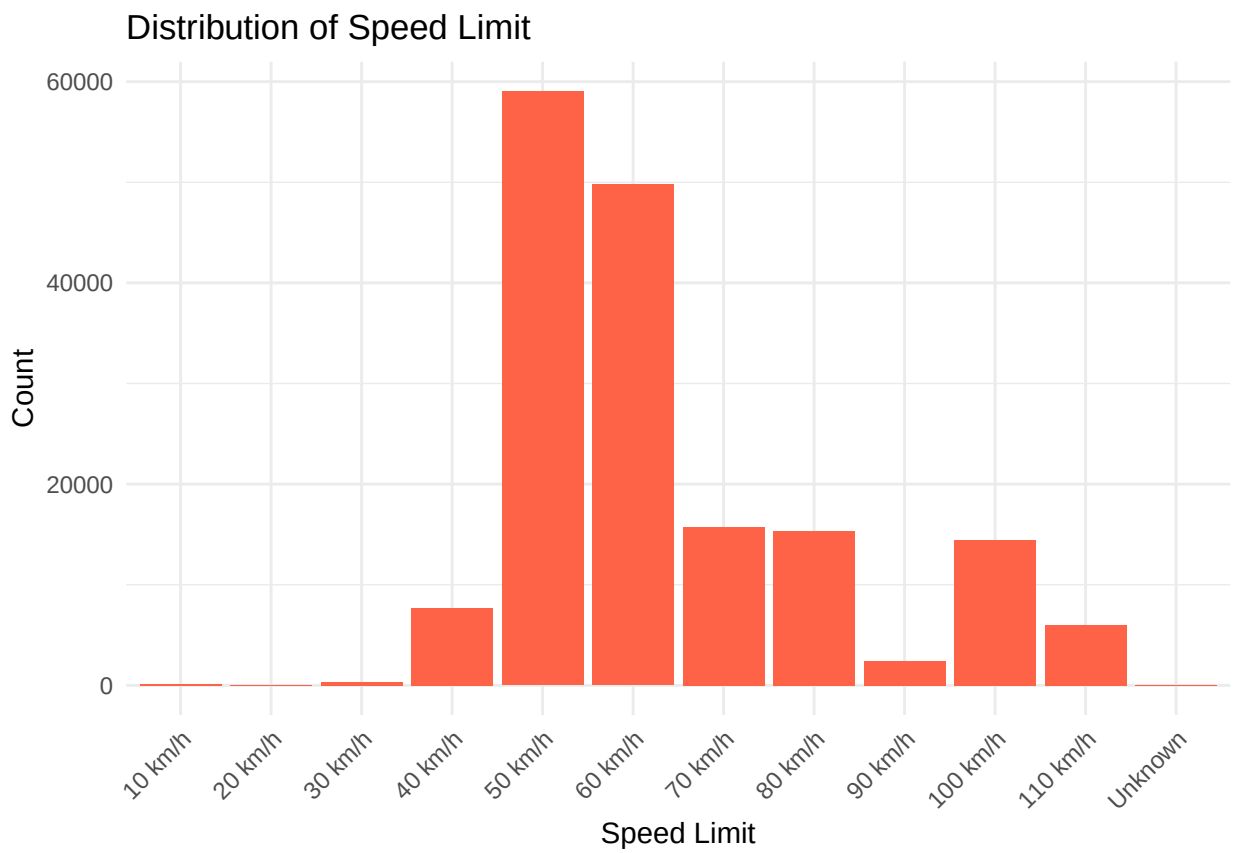

```r
ggplot(df_merged, aes(x = fct_infreq(Weather))) +
  geom_bar(fill = "steelblue") +
  labs(title = "Distribution of Weather Conditions",
       x = "Weather", y = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_y_continuous(labels = scales::comma)
```



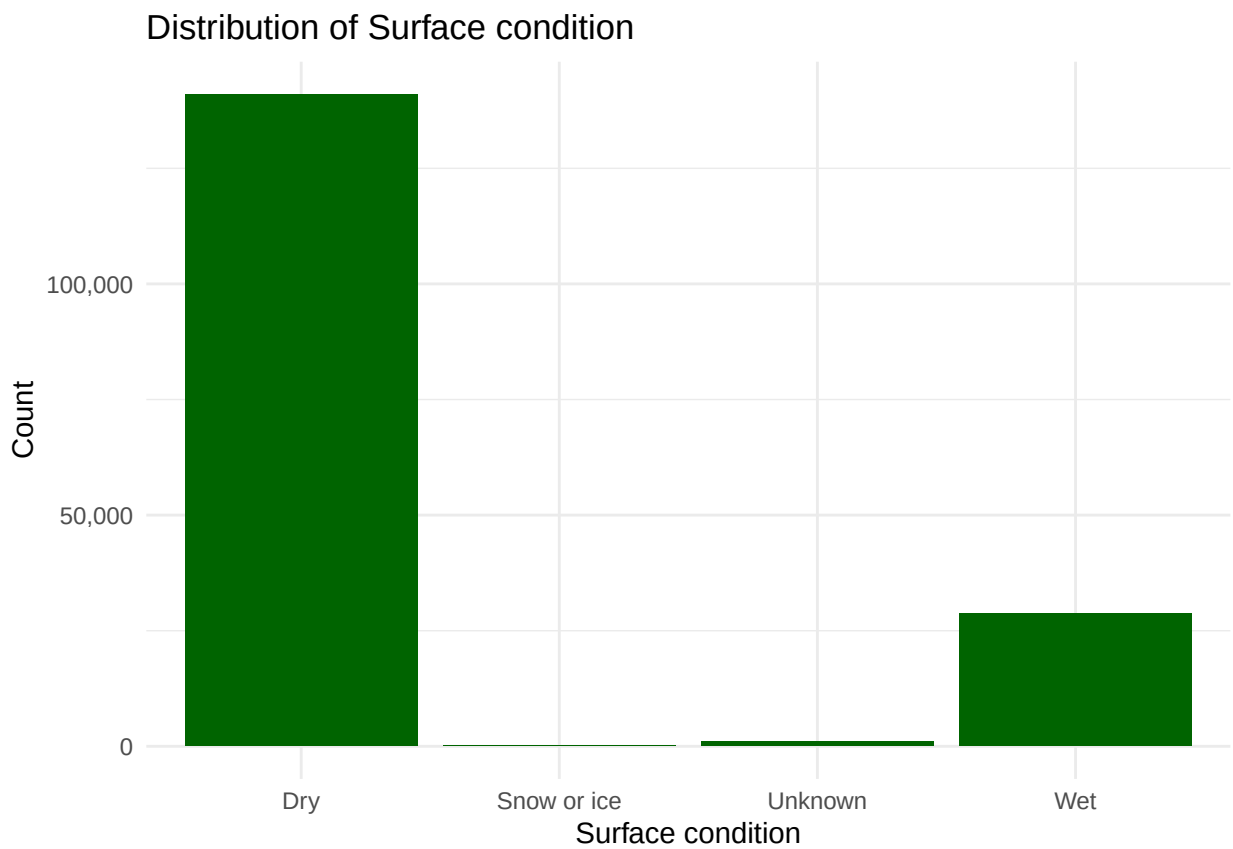

```
ggplot(df_merged, aes(x = fct_infreq(`Natural lighting`))) +
  geom_bar(fill = "orange") +
  labs(title = "Distribution of Natural Lighting", x = "Lighting", y = "Count") +
  theme_minimal()
```



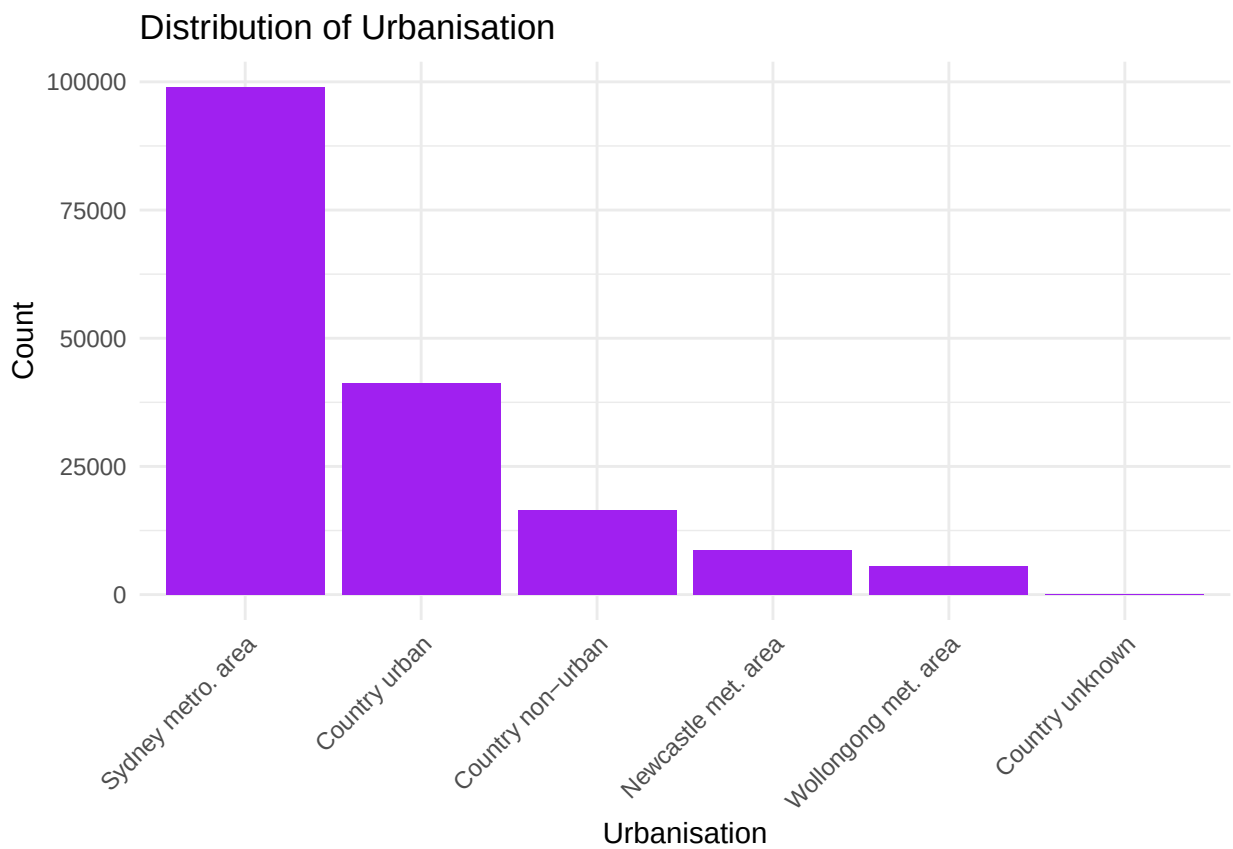

```r
df_merged$`Speed limit` <- factor(df_merged$`Speed limit`,
                                  levels = c("10 km/h", "20 km/h", "30 km/h", "40 km/h",
                                             "50 km/h", "60 km/h", "70 km/h", "80 km/h",
                                             "90 km/h", "100 km/h", "110 km/h", "Unknown"))

ggplot(df_merged, aes(x = `Speed limit`)) +
  geom_bar(fill = "tomato") +
  labs(title = "Distribution of Speed Limit", x = "Speed Limit", y = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



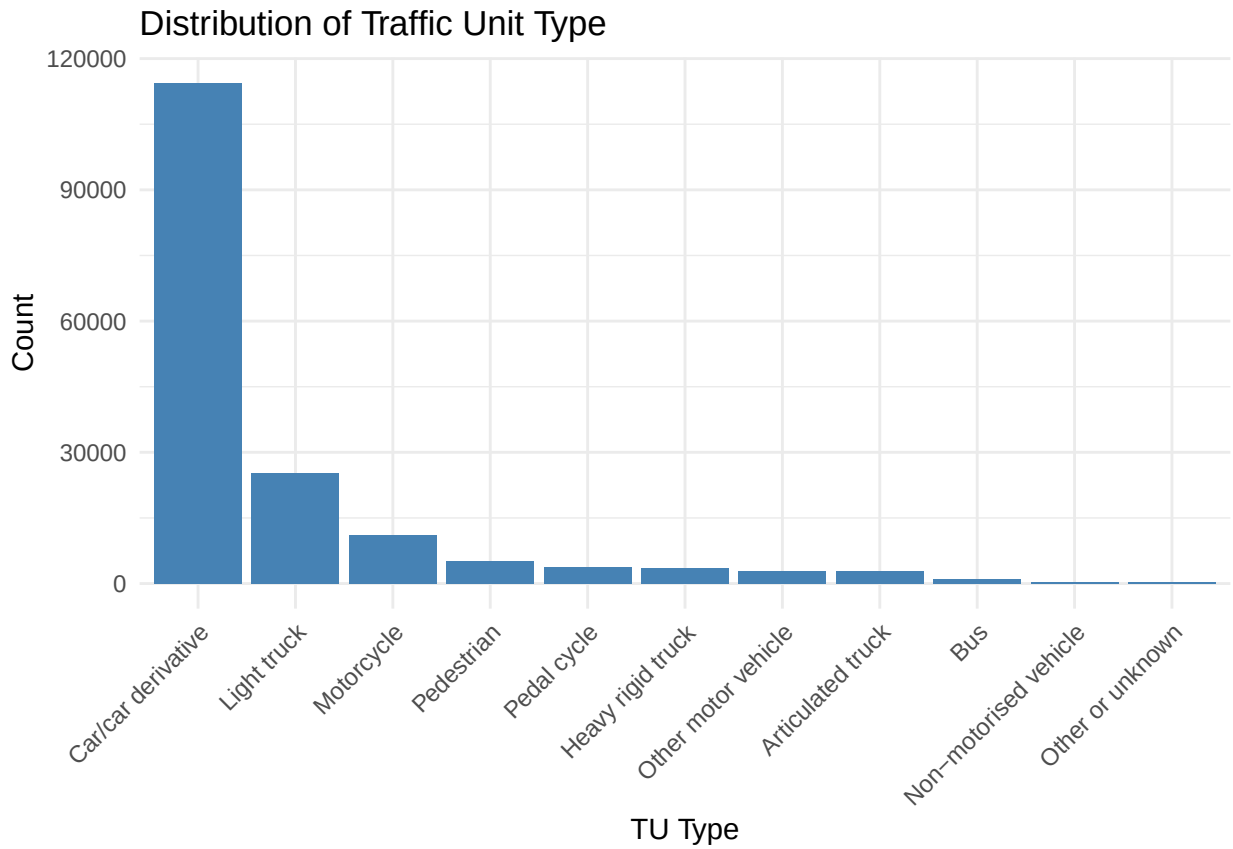

```
ggplot(df_merged, aes(x = `Surface condition`)) +
  geom_bar(fill = "darkgreen") +
  labs(title = "Distribution of Surface condition", x = "Surface condition", y = "Count") +
  theme_minimal()+
  scale_y_continuous(labels = scales::comma)
```



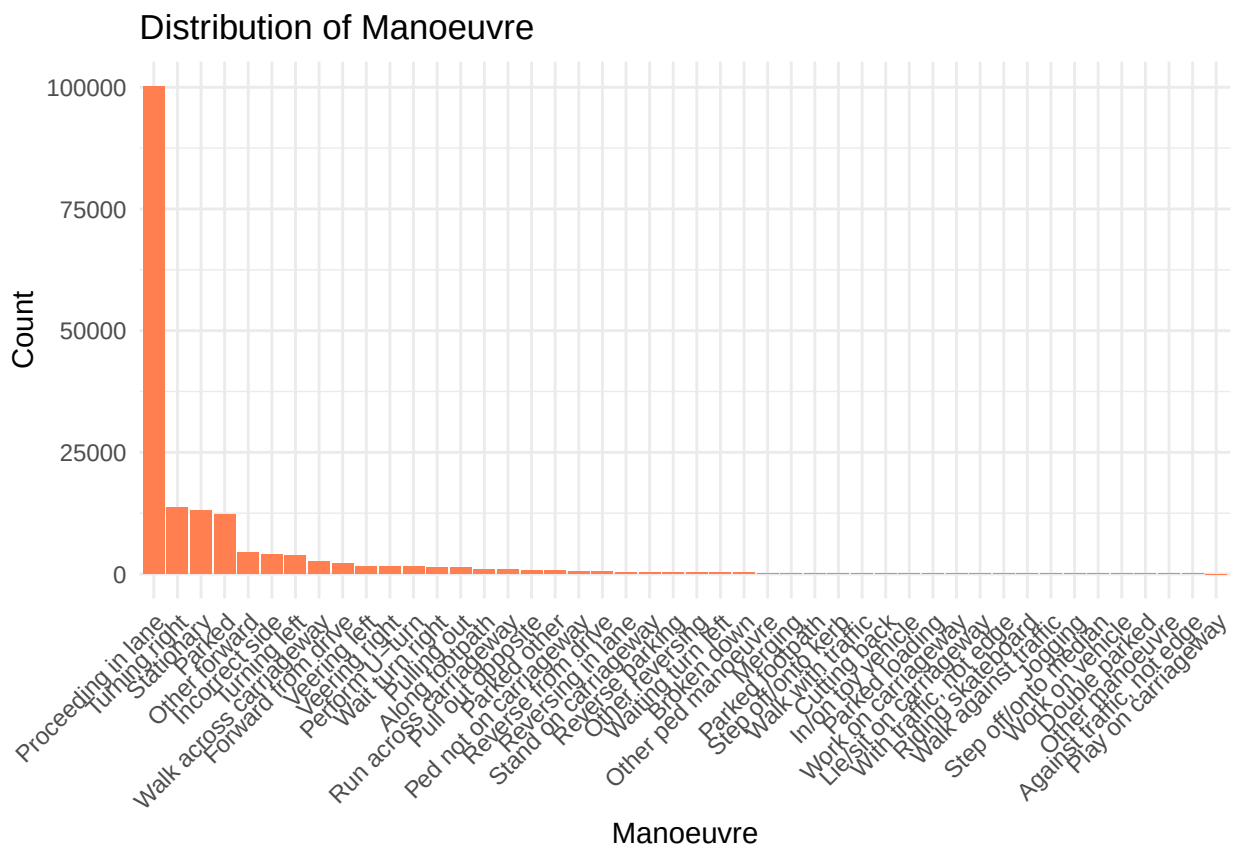

```
ggplot(df_merged, aes(x = fct_infreq(Urbanisation))) +
  geom_bar(fill = "purple") +
  labs(title = "Distribution of Urbanisation", x = "Urbanisation", y = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



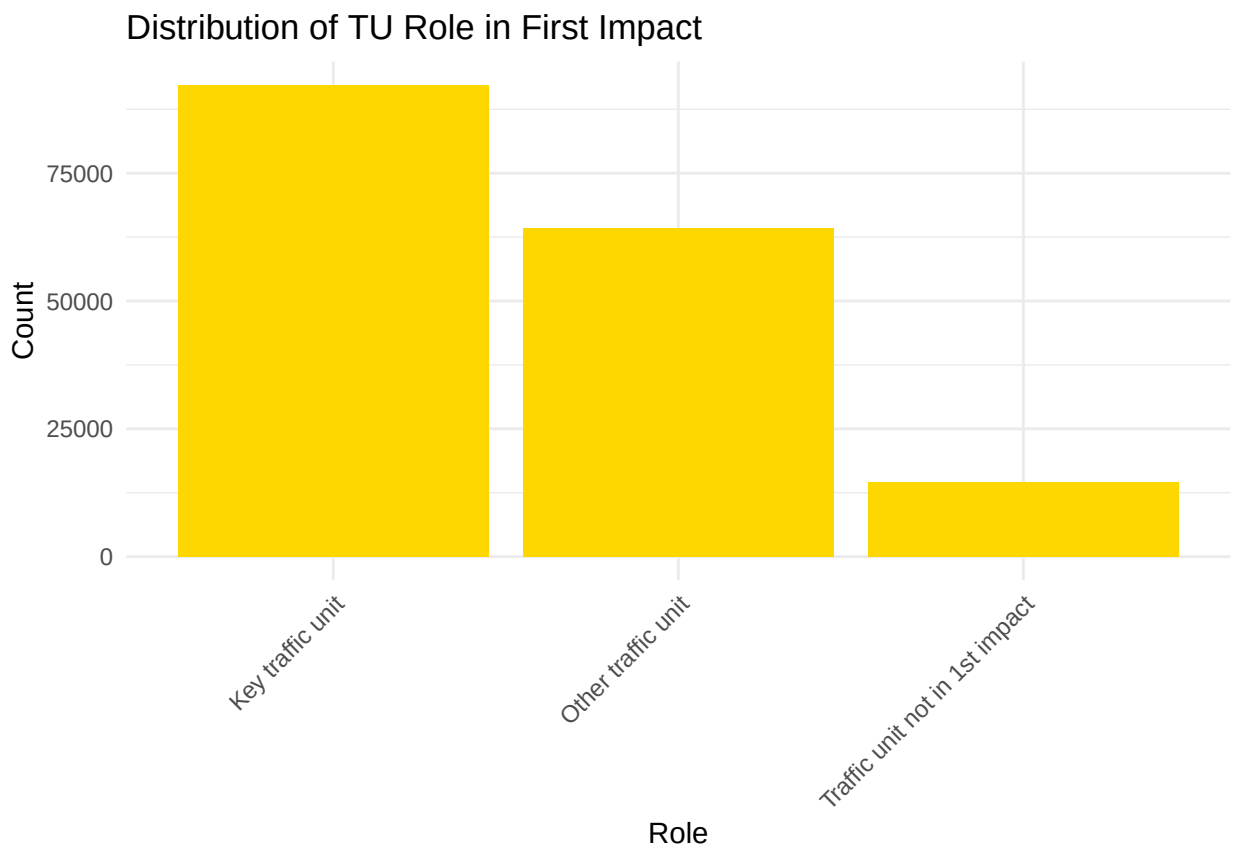

```r
ggplot(df_merged, aes(x = fct_infreq(`TU type group`))) +
    geom_bar(fill = "steelblue") +
    labs(title = "Distribution of Traffic Unit Type", x = "TU Type", y = "Count") +
    theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



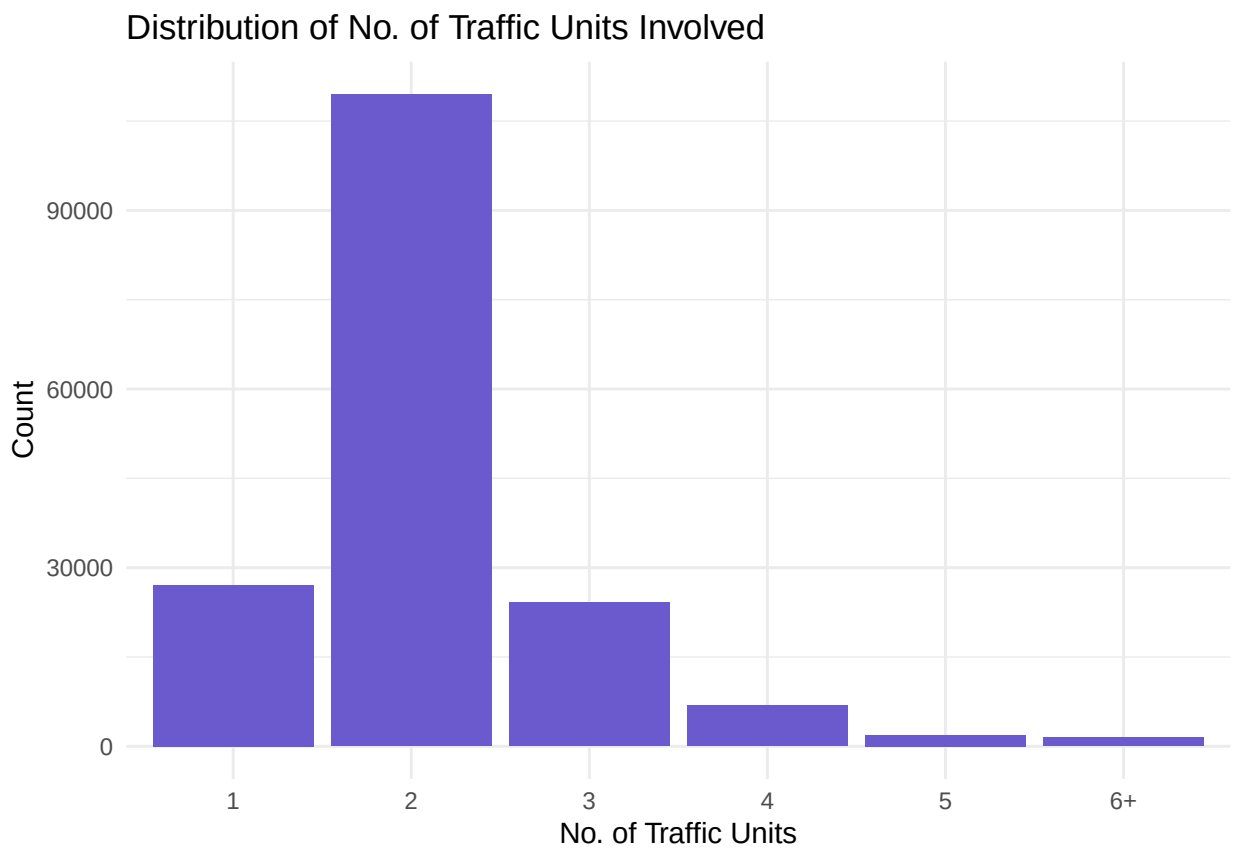

```r
ggplot(df_merged, aes(x = fct_infreq(Manoeuvre))) +
    geom_bar(fill = "coral") +
    labs(title = "Distribution of Manoeuvre", x = "Manoeuvre", y = "Count") +
    theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```r
ggplot(df_merged, aes(x = `TU role in first impact`)) +
    geom_bar(fill = "gold") +
    labs(title = "Distribution of TU Role in First Impact", x = "Role", y = "Count") +
    theme_minimal() +
    theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
df_tu <- df_merged %>%
    mutate(`TU group` = ifelse(`No. of traffic units involved` >= 6, "6+",
                               as.character(`No. of traffic units involved`)))
df_tu$`TU group` <- factor(df_tu$`TU group`, levels = c("1", "2", "3", "4", "5", "6+"))
ggplot(df_tu, aes(x = `TU group`)) +
geom_bar(fill = "slateblue") +
labs(title = "Distribution of No. of Traffic Units Involved", x = "No. of Traffic Units", y = "Count") +
```



The data shows significant class imbalance: Injury crashes are most frequent (114,355), followed by Non-casualty (54,191), with Fatal crashes being the minority (2,416). Most crashes occur in fine weather, daylight, and dry road conditions, suggesting traffic volume has more influence than weather on crash frequency. The 50-60 km/h speed zones account for the most crashes, with Sydney metro being the dominant urbanisation category. Car/car derivatives are the most common vehicle type, "Proceeding in lane" is the most frequent manoeuvre (~100,000), with all other manoeuvres being substantially less frequent. Key traffic units account for the largest share of roles in the first impact, and 2-vehicle crashes are the most common collision type.

3.1.2 Missing Values

```r
colSums(is.na(df_merged))
```

```
##                      Crash ID               Degree of crash
##                             0                             0
##    Degree of crash - detailed                Reporting year
##                             0                             0
##                 Year of crash                Month of crash
##                             0                             0
##          Day of week of crash            Two-hour intervals
##                             0                             0
##               Street of crash                   Street type
##                             0                             0
##                      Distance                     Direction
##                             0                             0
```

```
##           Identifying feature      Identifying feature type
##                             3                             0
##                          Town                     Route no.
##                             0                         62005
##          School zone location            School zone active
##                             0                             0
##              Type of location                      Latitude
##                             0                             0
##                     Longitude                           LGA
##                             0                             0
##                  Urbanisation                 Conurbation 1
##                             0                             0
##                     Alignment     Primary permanent feature
##                             0                        128408
##     Primary temporary feature     Primary hazardous feature
##                        168406                        168803
##               Street lighting                  Road surface
##                             0                             0
##             Surface condition                       Weather
##                             0                             0
##              Natural lighting             Signals operation
##                             0                             0
##         Other traffic control                   Speed limit
##                             0                             0
##   Road classification (admin)                    RUM - code
##                             0                             0
##             RUM - description                    DCA - code
##                             0                             0
##             DCA - description                DCA supplement
##                             0                        156545
##             First impact type                   Key TU type
##                             0                             0
##                 Other TU type No. of traffic units involved
##                         29195                             0
##                    No. killed         No. seriously injured
##                             0                             0
##        No. moderately injured       No. minor-other injured
##                             0                             0
##               Traffic unit ID            TU controlled flag
##                             0                             0
##       TU role in first impact                 TU type group
##                             0                             0
##              Street of travel           Direction of travel
##                             0                             0
##                     Manoeuvre                  Object hit 1
##                             0                        145702
##                  Object hit 2
##                        165996
```

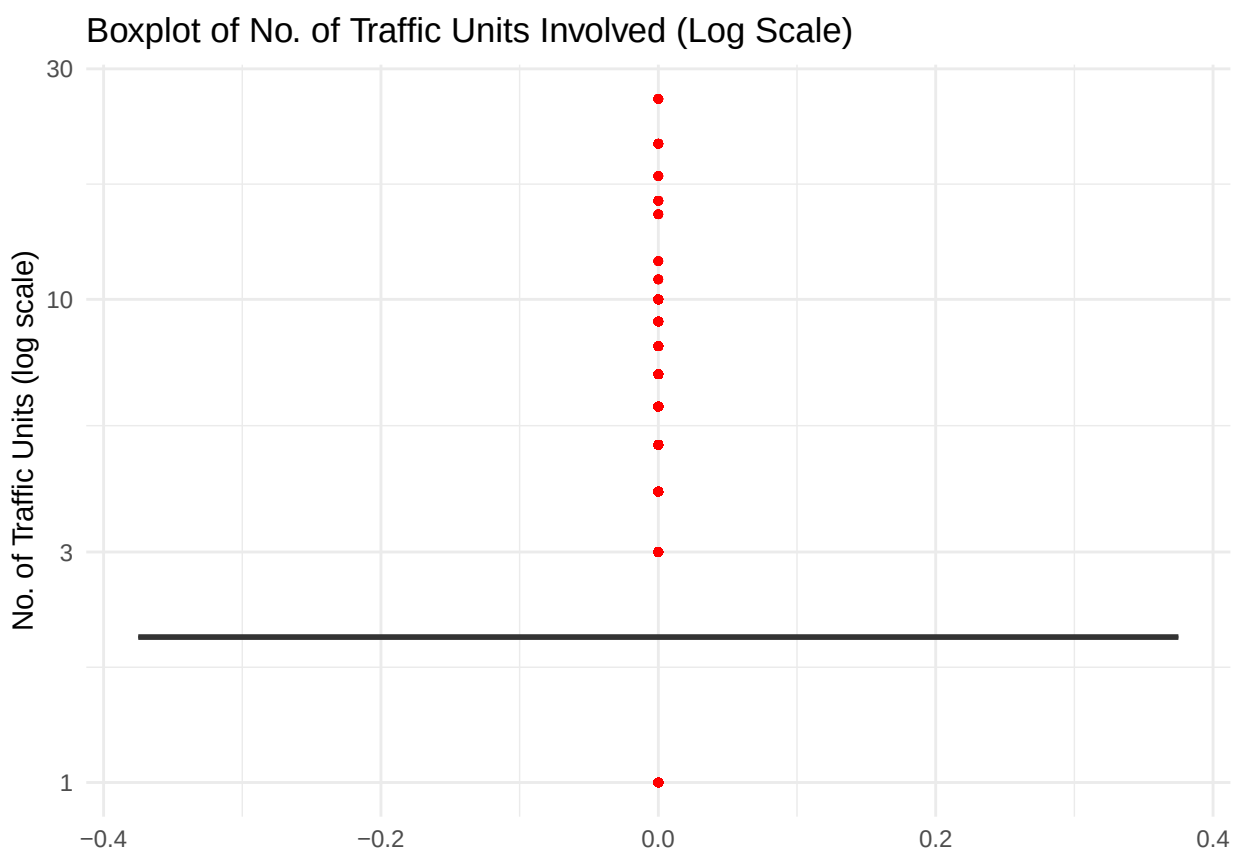
Core modelling variables (Weather, Speed limit, Natural lighting, Degree of crash) have no missing values. Primary hazardous feature (168,803), Primary temporary feature (168,406), and Object hit 2 (165,996) have very high missingness due to their conditional nature and will be dropped during preprocessing.

3.1.3 Outliers for Key Variables

```
summary(df_merged$`No. of traffic units involved`)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.000   2.000   2.000   2.149   2.000  26.000
```

```
ggplot(df_merged, aes(y = `No. of traffic units involved`)) +
  geom_boxplot(fill = "steelblue", outlier.colour = "red", outlier.size = 1) +
  scale_y_log10() +
  labs(title = "Boxplot of No. of Traffic Units Involved (Log Scale)",
       y = "No. of Traffic Units (log scale)") +
  theme_minimal()
```

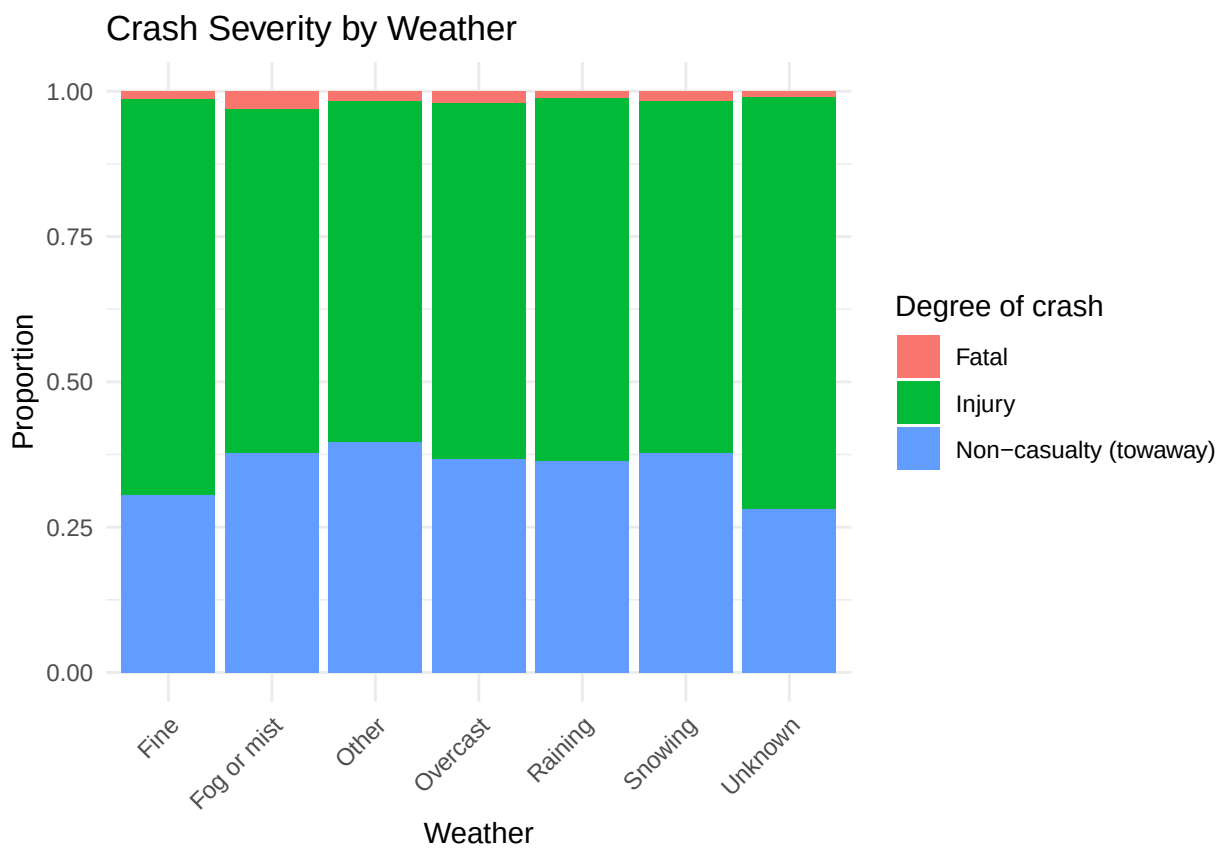


The number of traffic units has a median of 2 and mean of 2.149, with IQR concentrated between 1 and 2. The maximum value of 26 is an extreme outlier and may be capped during preprocessing.

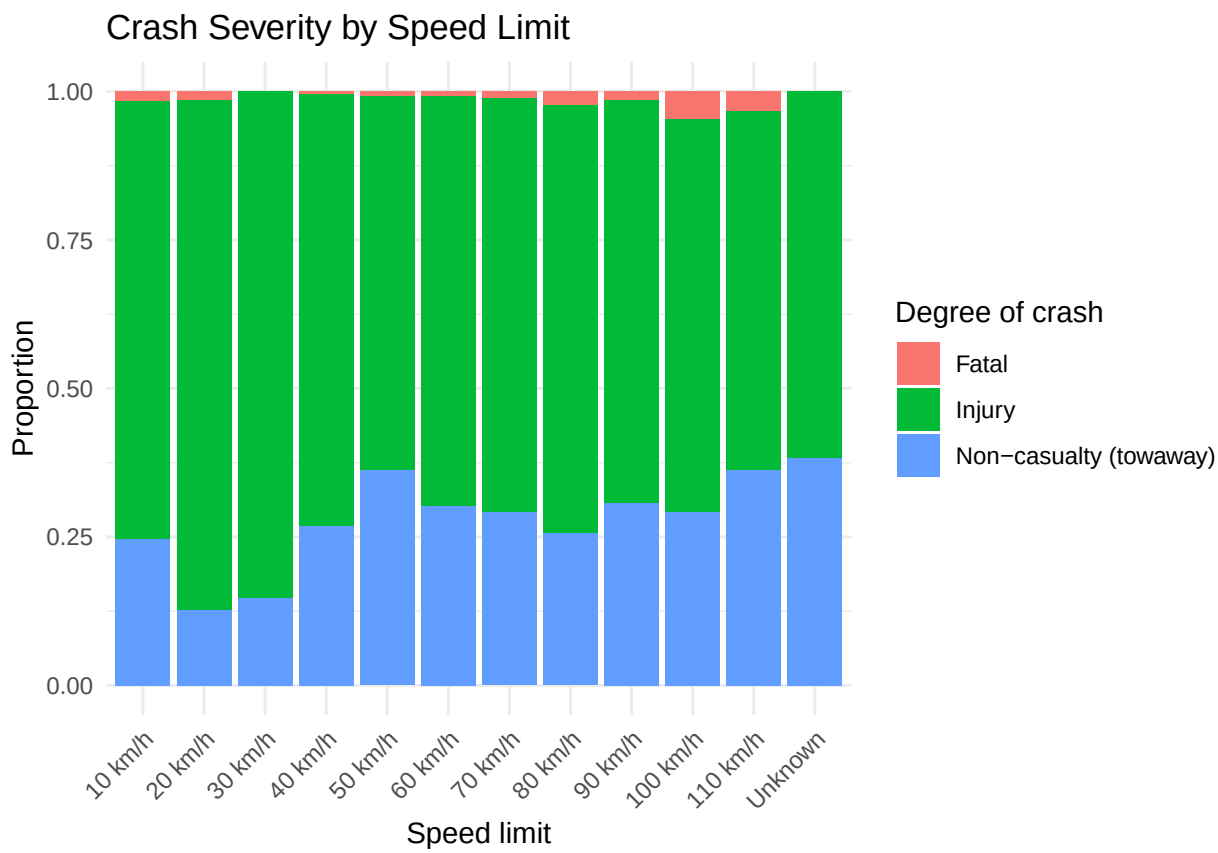
3.2 Bivariate Analysis

3.2.1 Features vs Outcomes

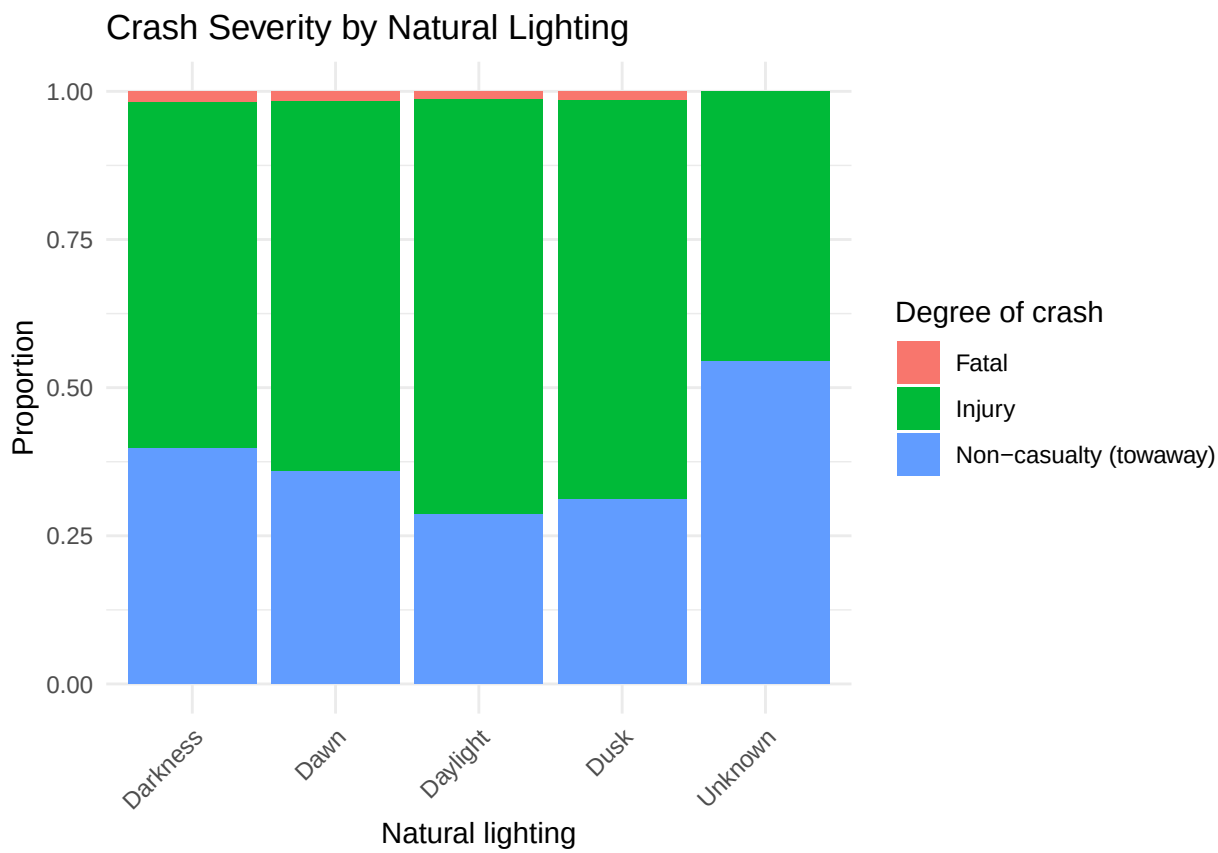
```r
ggplot(df_merged, aes(x = Weather, fill = `Degree of crash`)) +
  geom_bar(position = "fill") +
  labs(title = "Crash Severity by Weather", y = "Proportion") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



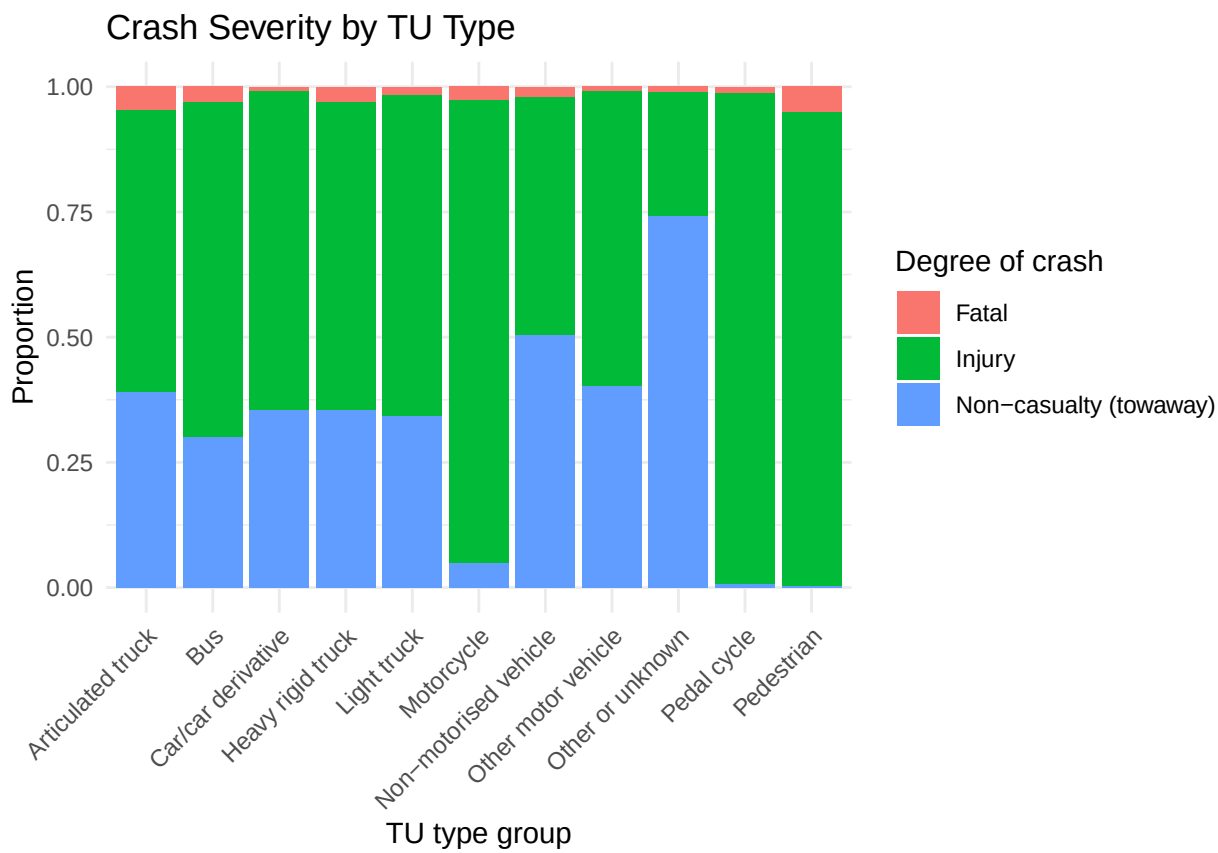

```r
ggplot(df_merged, aes(x = `Speed limit`, fill = `Degree of crash`)) +
  geom_bar(position = "fill") +
  labs(title = "Crash Severity by Speed Limit", y = "Proportion") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



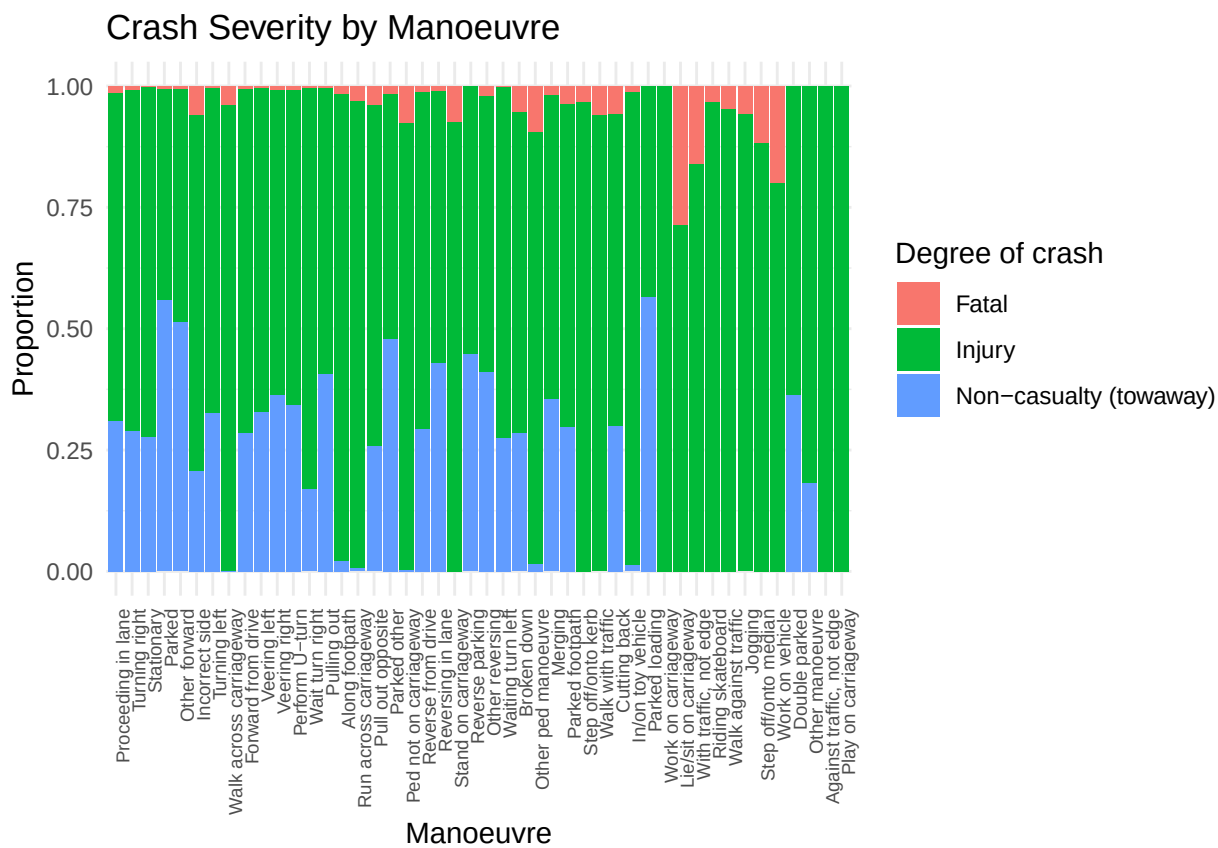

```r
ggplot(df_merged, aes(x = `Natural lighting`, fill = `Degree of crash`)) +
  geom_bar(position = "fill") +
  labs(title = "Crash Severity by Natural Lighting", y = "Proportion") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



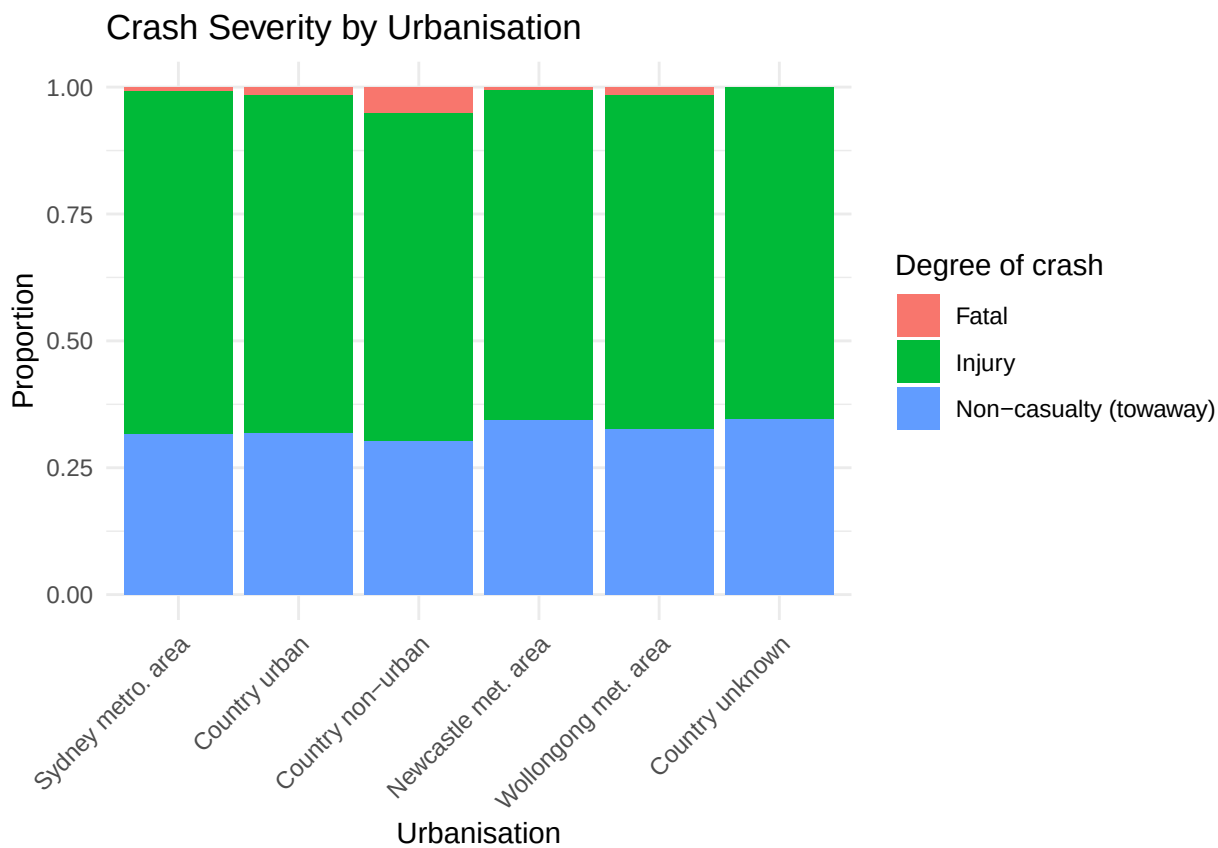

```r
ggplot(df_merged, aes(x = `TU type group`, fill = `Degree of crash`)) +
  geom_bar(position = "fill") +
  labs(title = "Crash Severity by TU Type", y = "Proportion") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



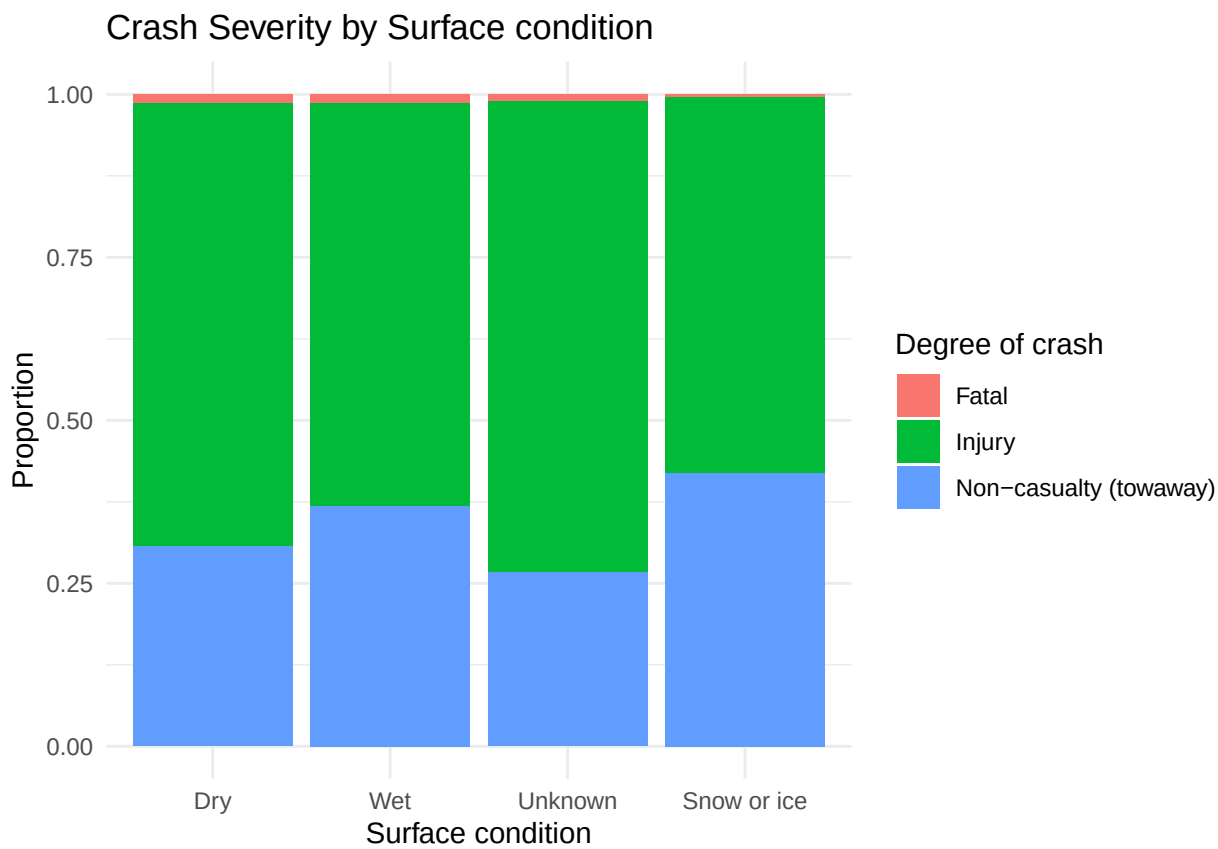

```r
ggplot(df_merged, aes(x = fct_infreq(factor(Manoeuvre)), fill = `Degree of crash`)) +
  geom_bar(position = "fill") +
  labs(title = "Crash Severity by Manoeuvre", x = "Manoeuvre", y = "Proportion") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, size = 7))
```



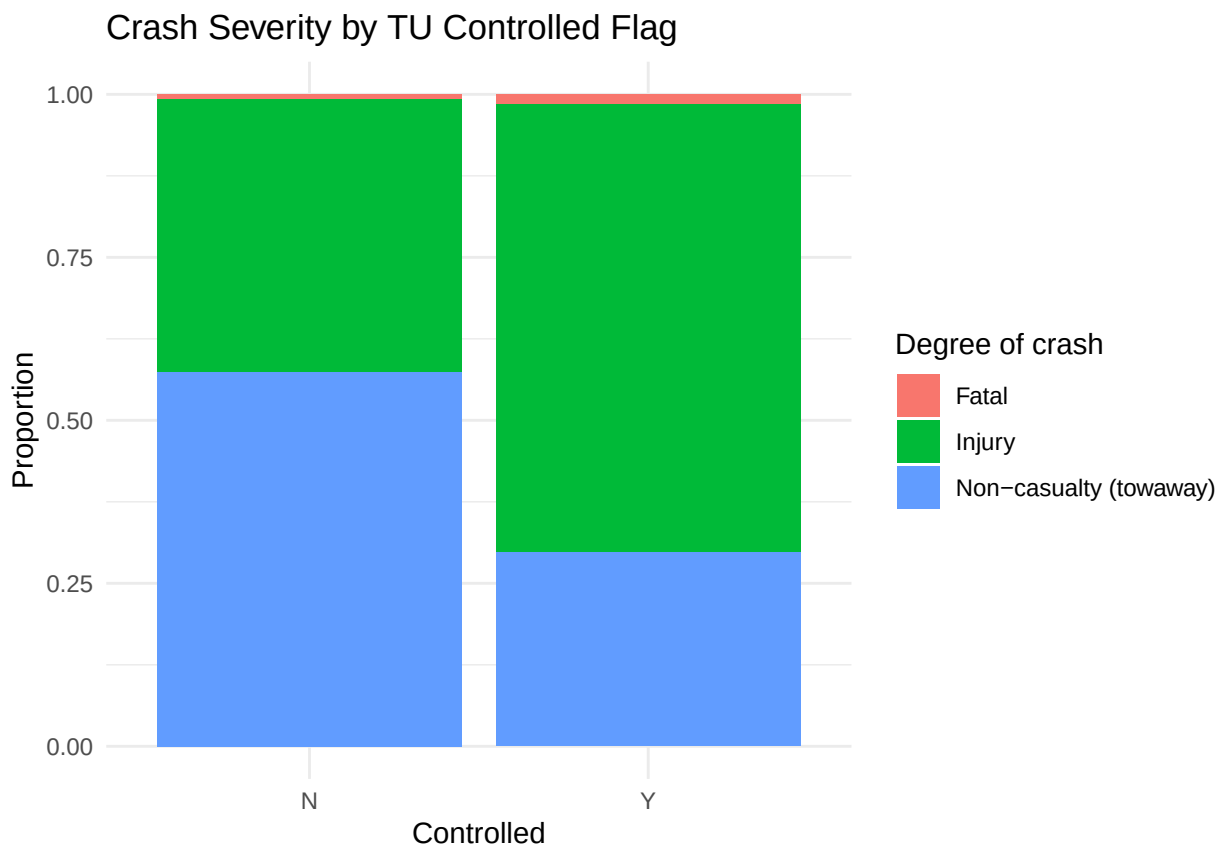

```r
ggplot(df_merged, aes(x = fct_infreq(factor(Urbanisation)), fill = `Degree of crash`)) +
  geom_bar(position = "fill") +
  labs(title = "Crash Severity by Urbanisation", x = "Urbanisation", y = "Proportion") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```




```
ggplot(df_merged, aes(x = fct_infreq(factor(`Surface condition`)), fill = `Degree of crash`)) +
  geom_bar(position = "fill") +
  labs(title = "Crash Severity by Surface condition", x = "Surface condition", y = "Proportion") +
  theme_minimal()
```



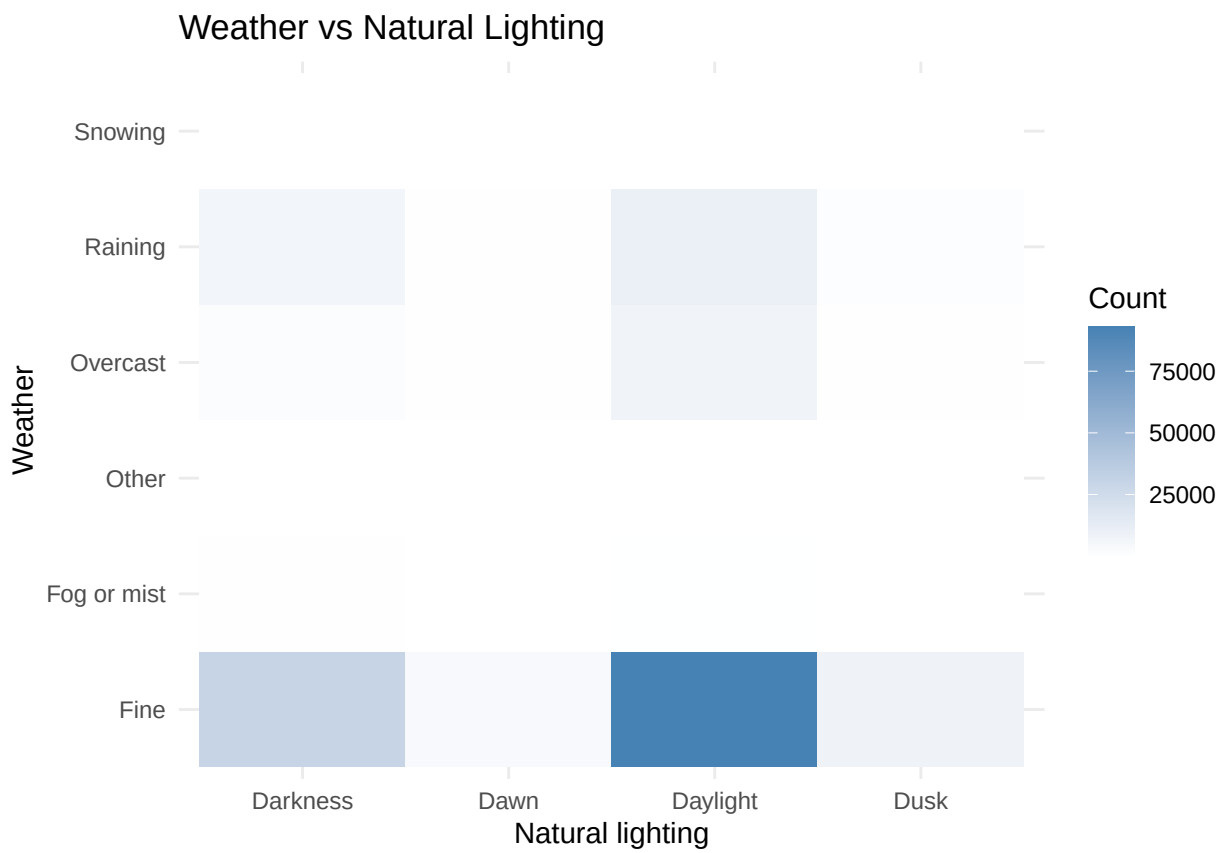

```
ggplot(df_merged, aes(x = `TU controlled flag`, fill = `Degree of crash`)) +
  geom_bar(position = "fill") +
  labs(title = "Crash Severity by TU Controlled Flag", x = "Controlled", y = "Proportion") +
  theme_minimal()
```



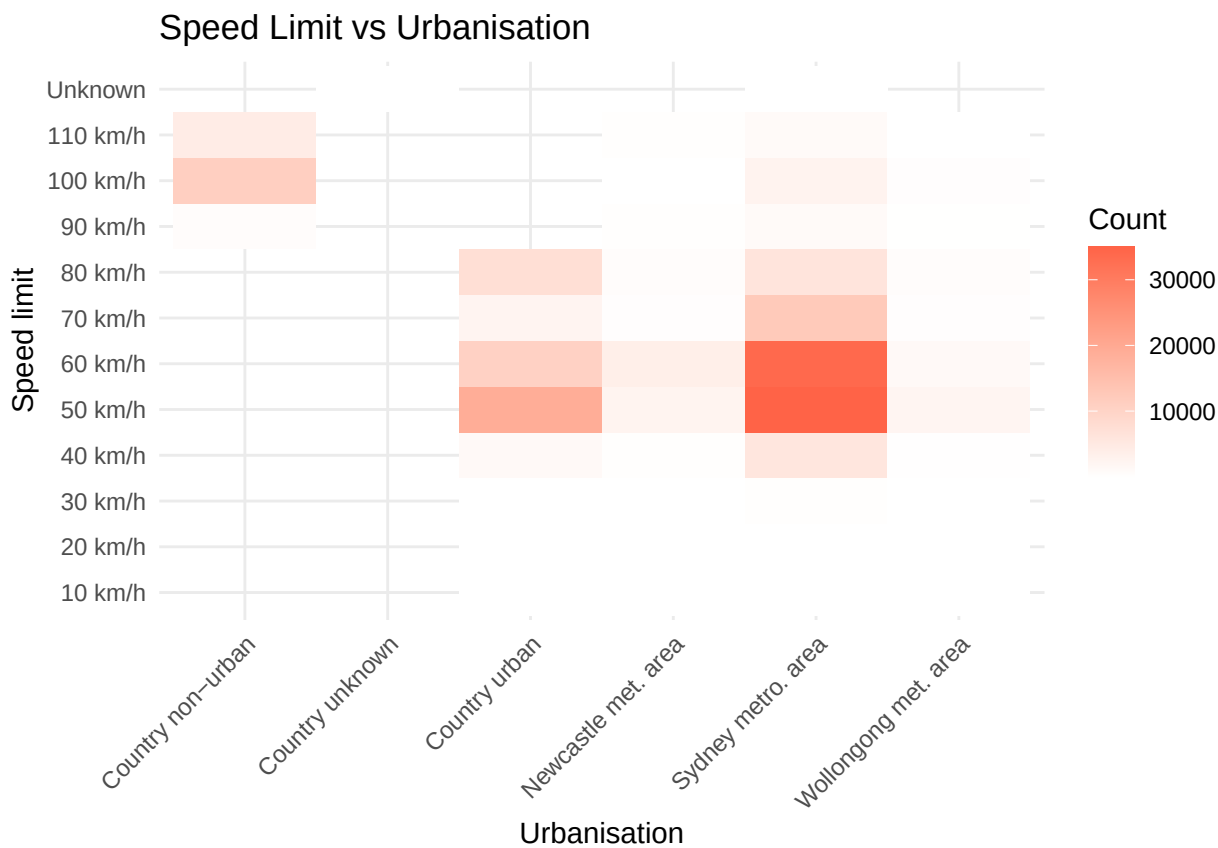
Speed limit shows the strongest relationship with crash severity, with higher speed zones (100-110 km/h) having noticeably greater Fatal proportions. Pedestrians and motorcycles show elevated Injury proportions, while uncontrolled traffic units have higher Non-casualty proportions. Country non-urban areas show slightly higher Fatal proportions compared to urban areas, consistent with higher rural speed limits.

3.2.2 Features vs Features

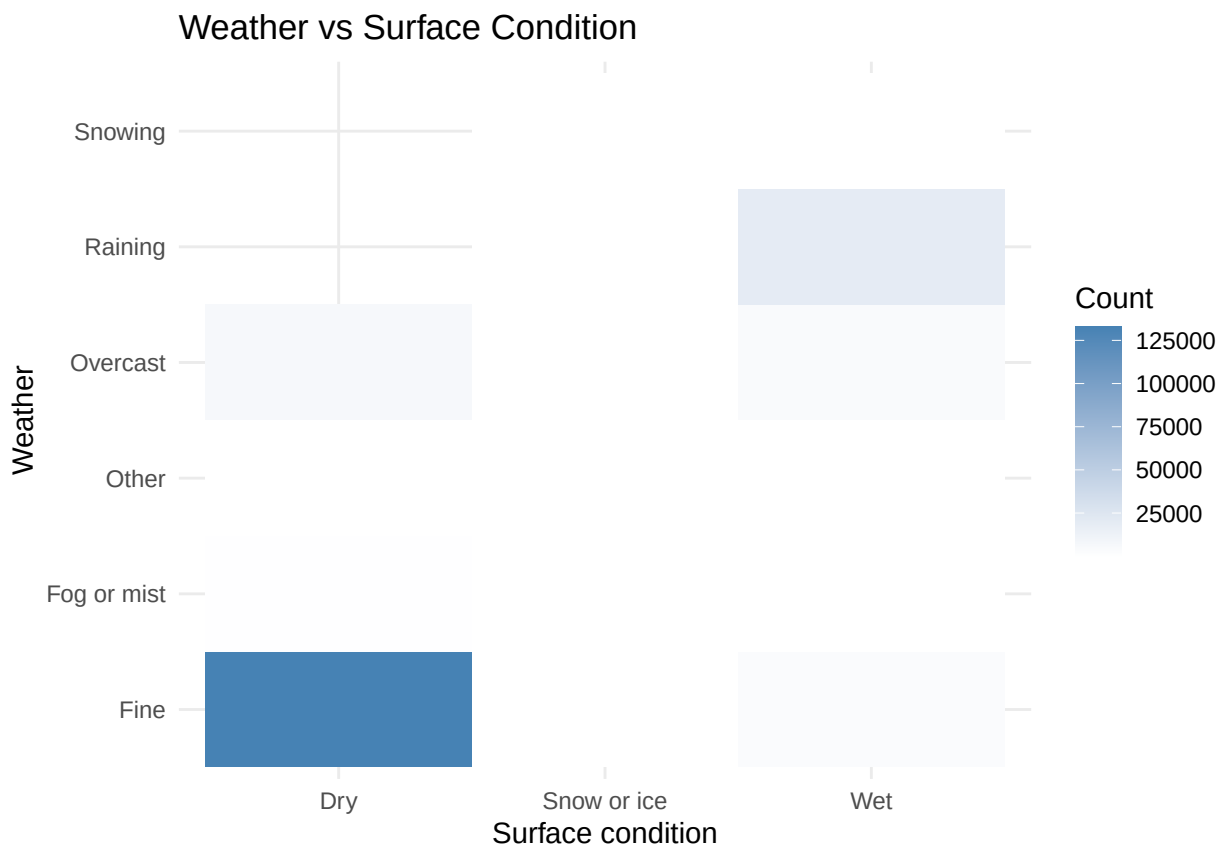
```
df_merged %>%
  group_by(Weather, `Natural lighting`) %>%
  summarise(count = n(), .groups = "drop") %>%
  filter(Weather != "Unknown", `Natural lighting` != "Unknown") %>%
  ggplot(aes(x = `Natural lighting`, y = Weather, fill = count)) +
  geom_tile() +
  scale_fill_gradient(low = "white", high = "steelblue") +
  labs(title = "Weather vs Natural Lighting", fill = "Count") +
  theme_minimal()
```



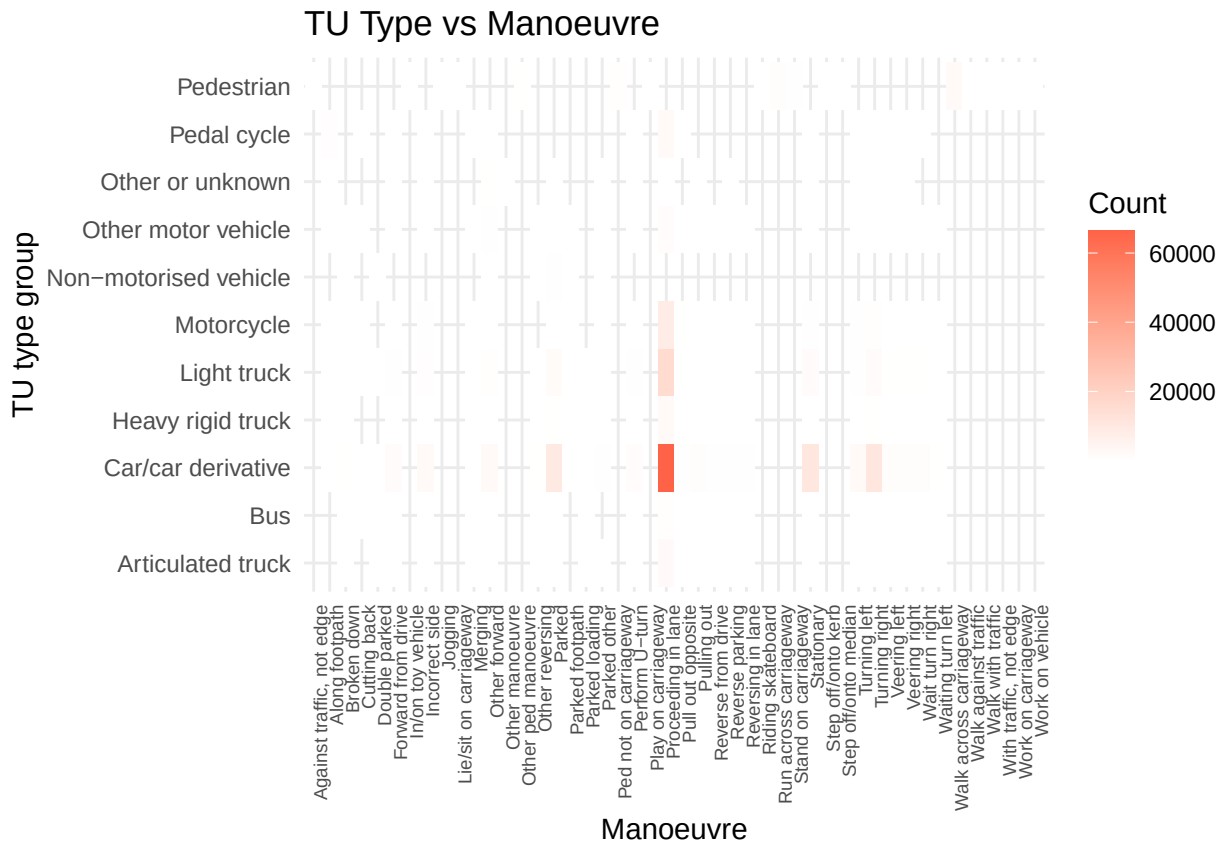

```r
df_merged %>%
  group_by(`Speed limit`, Urbanisation) %>%
  summarise(count = n(), .groups = "drop") %>%
  ggplot(aes(x = Urbanisation, y = `Speed limit`, fill = count)) +
  geom_tile() +
  scale_fill_gradient(low = "white", high = "tomato") +
  labs(title = "Speed Limit vs Urbanisation", fill = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



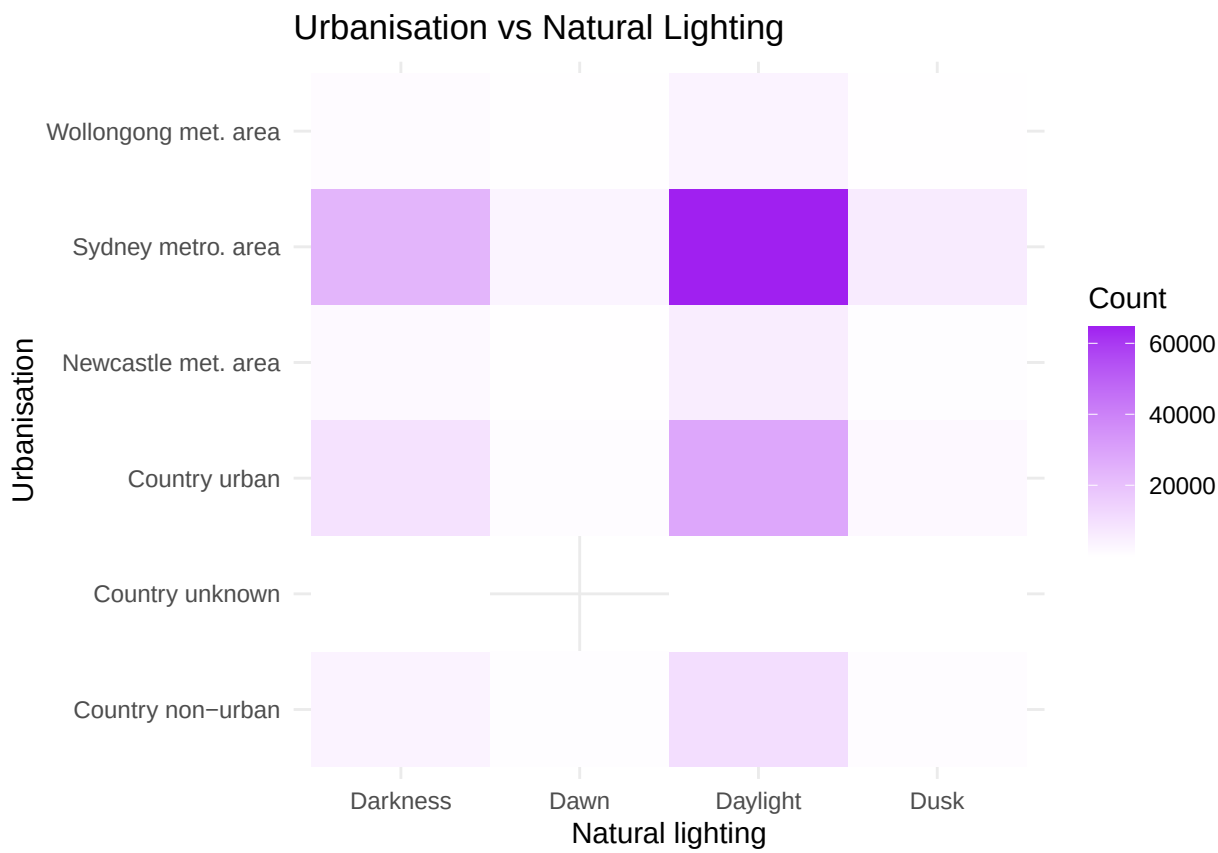

```r
df_merged %>%
  group_by(Weather, `Surface condition`) %>%
  summarise(count = n(), .groups = "drop") %>%
  filter(Weather != "Unknown", `Surface condition` != "Unknown") %>%
  ggplot(aes(x = `Surface condition`, y = Weather, fill = count)) +
  geom_tile() +
  scale_fill_gradient(low = "white", high = "steelblue") +
  labs(title = "Weather vs Surface Condition", fill = "Count") +
  theme_minimal()
```




```r
df_merged %>%
  group_by(`TU type group`, Manoeuvre) %>%
  summarise(count = n(), .groups = "drop") %>%
  ggplot(aes(x = Manoeuvre, y = `TU type group`, fill = count)) +
  geom_tile() +
  scale_fill_gradient(low = "white", high = "tomato") +
  labs(title = "TU Type vs Manoeuvre", fill = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, size = 7))
```




```
df_merged %>%
  group_by(Urbanisation, `Natural lighting`) %>%
  summarise(count = n(), .groups = "drop") %>%
  filter(`Natural lighting` != "Unknown") %>%
  ggplot(aes(x = `Natural lighting`, y = Urbanisation, fill = count)) +
  geom_tile() +
  scale_fill_gradient(low = "white", high = "purple") +
  labs(title = "Urbanisation vs Natural Lighting", fill = "Count") +
  theme_minimal()
```



Fine weather strongly aligns with dry surfaces while rain almost exclusively pairs with wet conditions, confirming dataset consistency. Speed limits of 50-60 km/h dominate Sydney metro areas while 100-110 km/h zones concentrate in Country non-urban regions. Crashes predominantly occur in daylight across all urbanisation categories, with Sydney metro having the highest overall crash volume.

3.3 Data Visualisation: Additional Graphs/Tables with Insights

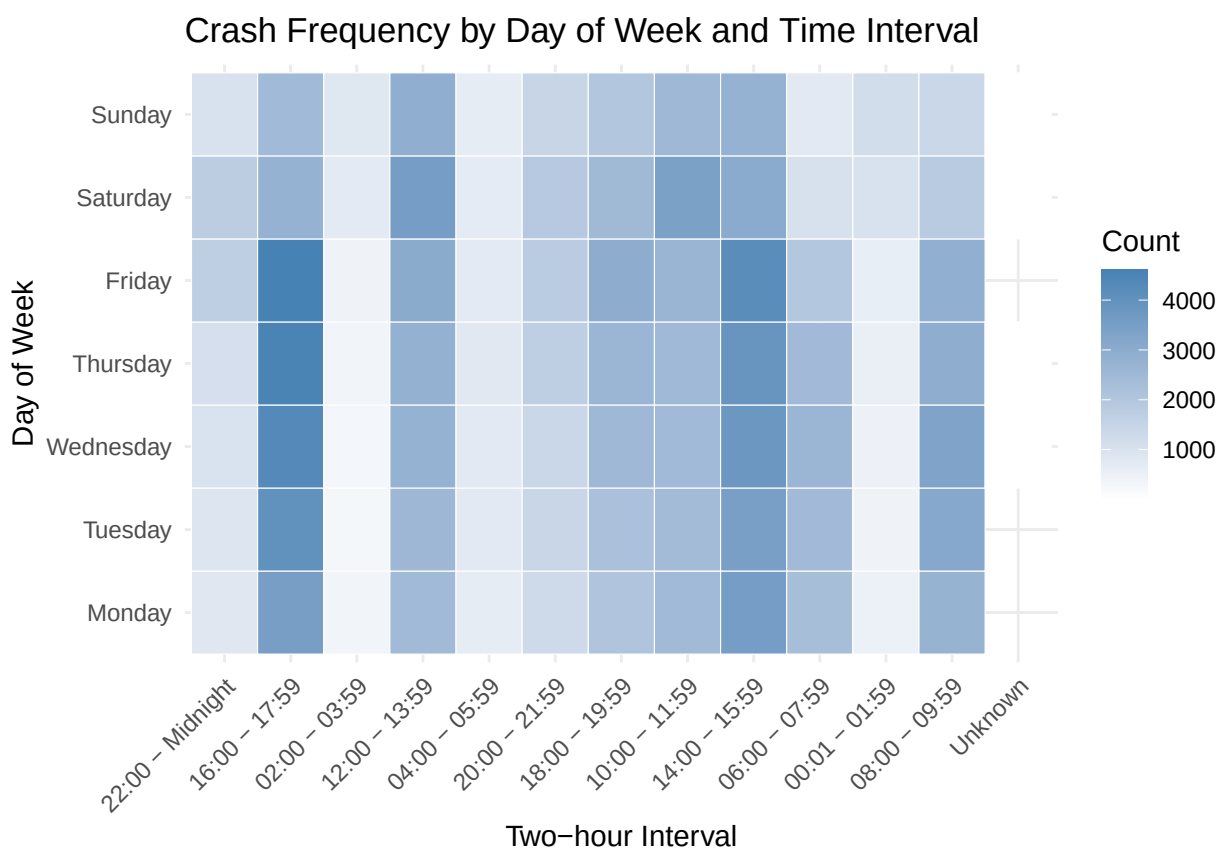
To further support the exploratory analysis, additional visualizations and summary tables were produced to reveal temporal, spatial, and contextual patterns in crash severity. These displays go beyond simple univariate and bivariate summaries by highlighting high-risk combinations of conditions and identifying patterns that may influence feature selection and model choice.

3.3.1 Heatmap-day-time

```
library(knitr)

# Ensure ordered factors for time-related variables
day_levels <- c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday")
df_merged$`Day of week of crash` <- factor(df_merged$`Day of week of crash`, levels = day_levels)

# If Two-hour intervals is character, keep original order as it appears
df_merged$`Two-hour intervals` <- factor(df_merged$`Two-hour intervals`,
                                         levels = unique(df_merged$`Two-hour intervals`))
```

```r
# Order reporting year if needed
df_merged$`Reporting year` <- factor(df_merged$`Reporting year`,
                                     levels = sort(unique(df_merged$`Reporting year`)))
```

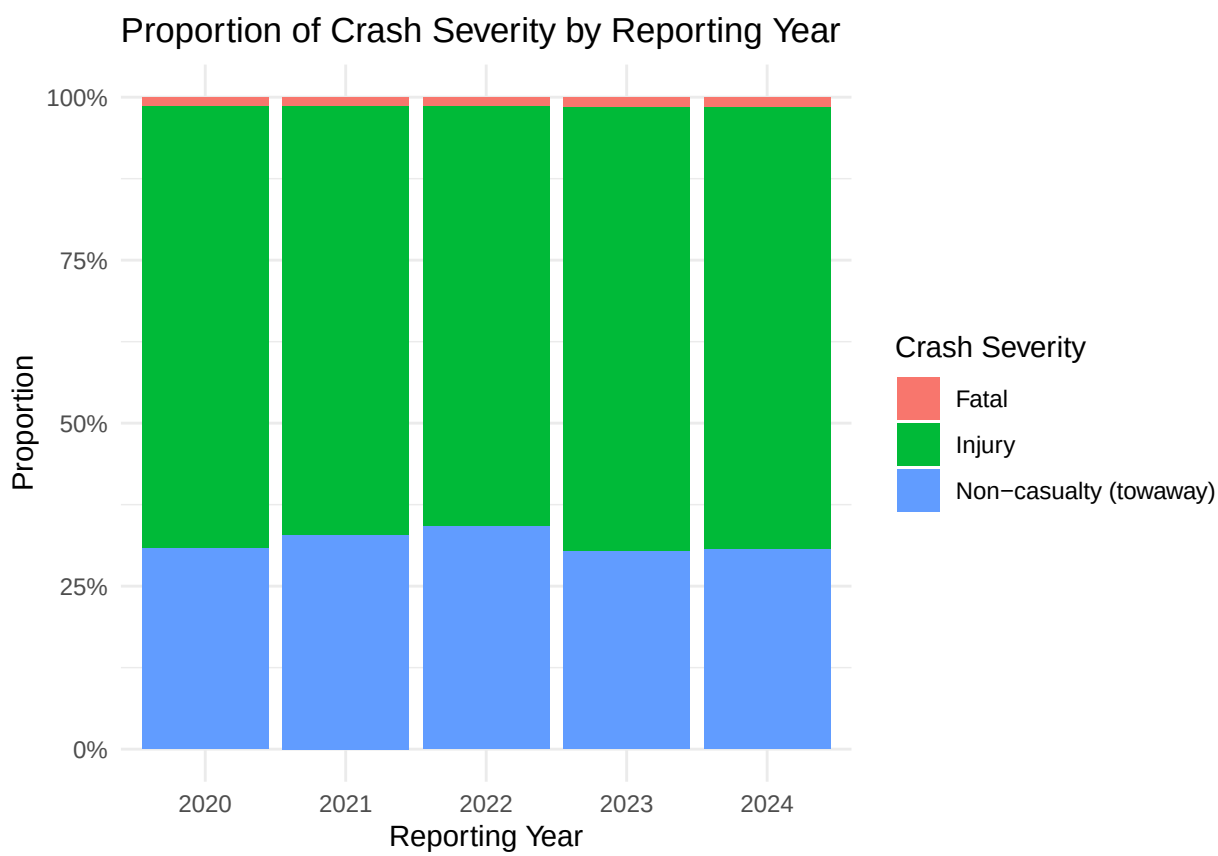
```r
df_merged %>%
  group_by(`Day of week of crash`, `Two-hour intervals`) %>%
  summarise(count = n(), .groups = "drop") %>%
  ggplot(aes(x = `Two-hour intervals`, y = `Day of week of crash`, fill = count)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "white", high = "steelblue") +
  labs(
    title = "Crash Frequency by Day of Week and Time Interval",
    x = "Two-hour Interval",
    y = "Day of Week",
    fill = "Count"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



The heatmap shows that crash frequency is concentrated in specific time periods rather than being evenly distributed across the week. In particular, daytime and peak commuting periods tend to have the highest crash counts, suggesting that traffic exposure and congestion may play a major role in crash occurrence. This pattern indicates that time-related variables are likely to be useful predictors in the classification task.

3.3.2 Severity-by-year

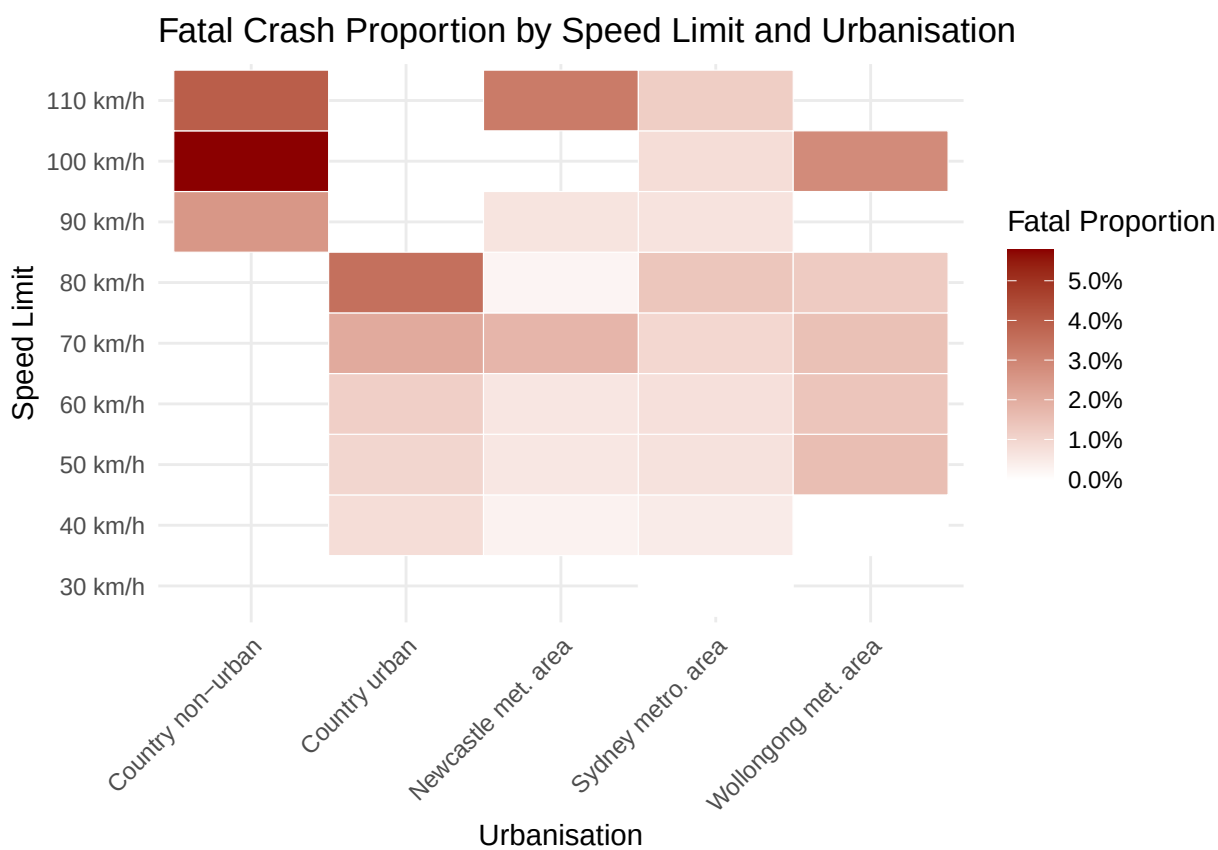
```
df_merged %>%
  group_by(`Reporting year`, `Degree of crash`) %>%
  summarise(count = n(), .groups = "drop") %>%
  ggplot(aes(x = `Reporting year`, y = count, fill = `Degree of crash`)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(
    title = "Proportion of Crash Severity by Reporting Year",
    x = "Reporting Year",
    y = "Proportion",
    fill = "Crash Severity"
  ) +
  theme_minimal()
```



The proportional distribution of crash severity appears relatively stable across reporting years, with Injury crashes consistently dominating the dataset and Fatal crashes remaining the minority class. This suggests that class imbalance is a persistent characteristic of the data rather than a one-off issue in a single year, reinforcing the need for imbalance-aware modelling and evaluation.

3.3.3 Fatal-rate-speed-urban

```r
fatal_heatmap <- df_merged %>%
  filter(!is.na(`Speed limit`), !is.na(Urbanisation), !is.na(`Degree of crash`)) %>%
  group_by(`Speed limit`, Urbanisation) %>%
  summarise(
    total = n(),
    fatal_rate = mean(`Degree of crash` == "Fatal"),
    .groups = "drop"
  ) %>%
  filter(total >= 100)  # avoid unstable rates from tiny groups

ggplot(fatal_heatmap, aes(x = Urbanisation, y = `Speed limit`, fill = fatal_rate)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "white", high = "darkred",
                      labels = scales::percent_format(accuracy = 0.1)) +
  labs(
    title = "Fatal Crash Proportion by Speed Limit and Urbanisation",
    x = "Urbanisation",
    y = "Speed Limit",
    fill = "Fatal Proportion"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



The heatmap suggests that the proportion of Fatal crashes is higher in high-speed areas, especially in less urbanised regions. This supports the earlier bivariate findings and indicates that speed environment and spatial context jointly influence crash severity. These interaction patterns justify the use of flexible models

such as decision trees, random forests, and boosting methods.

3.3.4 High-Risk Crash Scenarios

```
risk_table <- df_merged %>%
  filter(
    !is.na(`Speed limit`),
    !is.na(`Natural lighting`),
    !is.na(Urbanisation),
    !is.na(`Degree of crash`)
  ) %>%
  group_by(`Speed limit`, `Natural lighting`, Urbanisation) %>%
  summarise(
    total_crashes = n(),
    fatal_prop = mean(`Degree of crash` == "Fatal"),
    injury_prop = mean(`Degree of crash` == "Injury"),
    non_casualty_prop = mean(`Degree of crash` == "Non-casualty (towaway)"),
    .groups = "drop"
  ) %>%
  filter(total_crashes >= 200) %>%   # keep only meaningful groups
  arrange(desc(fatal_prop), desc(total_crashes)) %>%   # sort by fatal proportion descending
  slice_head(n = 10) %>%
  mutate(
    fatal_prop = scales::percent(fatal_prop, accuracy = 0.1),
    injury_prop = scales::percent(injury_prop, accuracy = 0.1),
    non_casualty_prop = scales::percent(non_casualty_prop, accuracy = 0.1)
  ) %>%
  rename(
    `Speed Limit` = `Speed limit`,
    `Natural Lighting` = `Natural lighting`,
    `Total Crashes` = total_crashes,
    `Fatal Proportion` = fatal_prop,
    `Injury Proportion` = injury_prop,
    `Non-casualty Proportion` = non_casualty_prop
  )

knitr::kable(
  risk_table,
  caption = "Table: High-risk crash scenarios ranked by fatal crash proportion"
)
```

Table 1: Table: High-risk crash scenarios ranked by fatal crash proportion

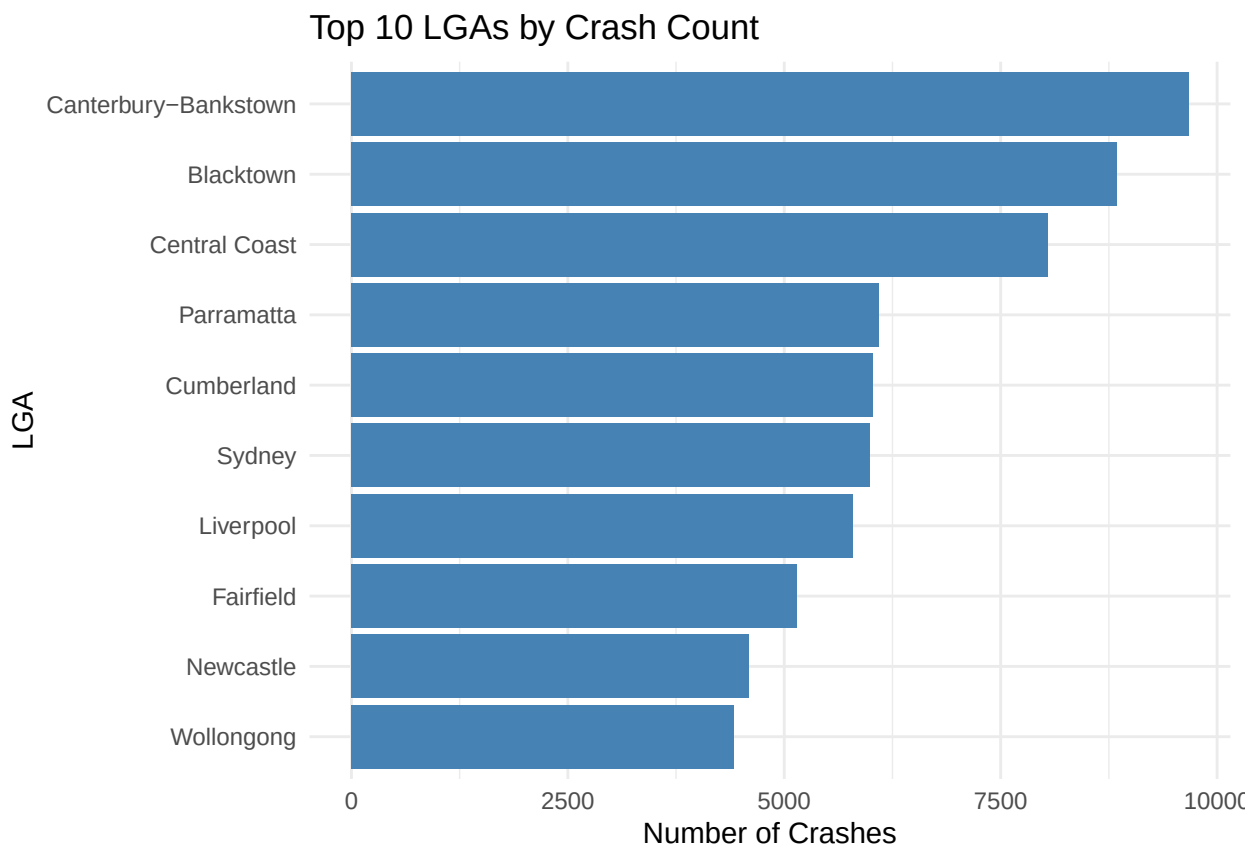
Speed Limit	Natural Lighting	Urbanisation	Total Crashes	Fatal Proportion	Injury Proportion	Non-casualty Proportion
110 km/h	Dusk	Country non-urban	230	7.8%	51.7%	40.4%
100 km/h	Darkness	Country non-urban	2565	6.8%	59.1%	34.1%

Speed Limit	Natural Lighting	Urbanisation	Total Crashes	Fatal Proportion	Injury Proportion	Non-casualty Proportion
100 km/h	Dusk	Country non-urban	626	6.7%	58.8%	34.5%
100 km/h	Daylight	Country non-urban	7490	5.5%	70.1%	24.3%
110 km/h	Darkness	Country non-urban	1215	5.4%	57.4%	37.1%
80 km/h	Dawn	Country urban	270	4.8%	66.3%	28.9%
70 km/h	Darkness	Country urban	463	4.8%	57.9%	37.4%
80 km/h	Darkness	Country urban	1536	4.7%	67.9%	27.4%
110 km/h	Darkness	Sydney metro. area	245	4.5%	59.6%	35.9%
100 km/h	Dawn	Country non-urban	455	4.2%	63.3%	32.5%

The table highlights combinations of environmental and contextual conditions associated with relatively high fatal proportions. Rather than relying on a single variable in isolation, these results suggest that crash severity is shaped by combinations of speed environment, lighting conditions, and urban context. This motivates the inclusion of interaction-capable models in the modelling phase.

3.3.5 Top 10 LGA by crash count

```
top_lga <- df_merged %>%
  filter(!is.na(LGA)) %>%
  count(LGA, sort = TRUE) %>%
  slice_head(n = 10)

ggplot(top_lga, aes(x = reorder(LGA, n), y = n)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(
    title = "Top 10 LGAs by Crash Count",
    x = "LGA",
    y = "Number of Crashes"
  ) +
  theme_minimal()
```

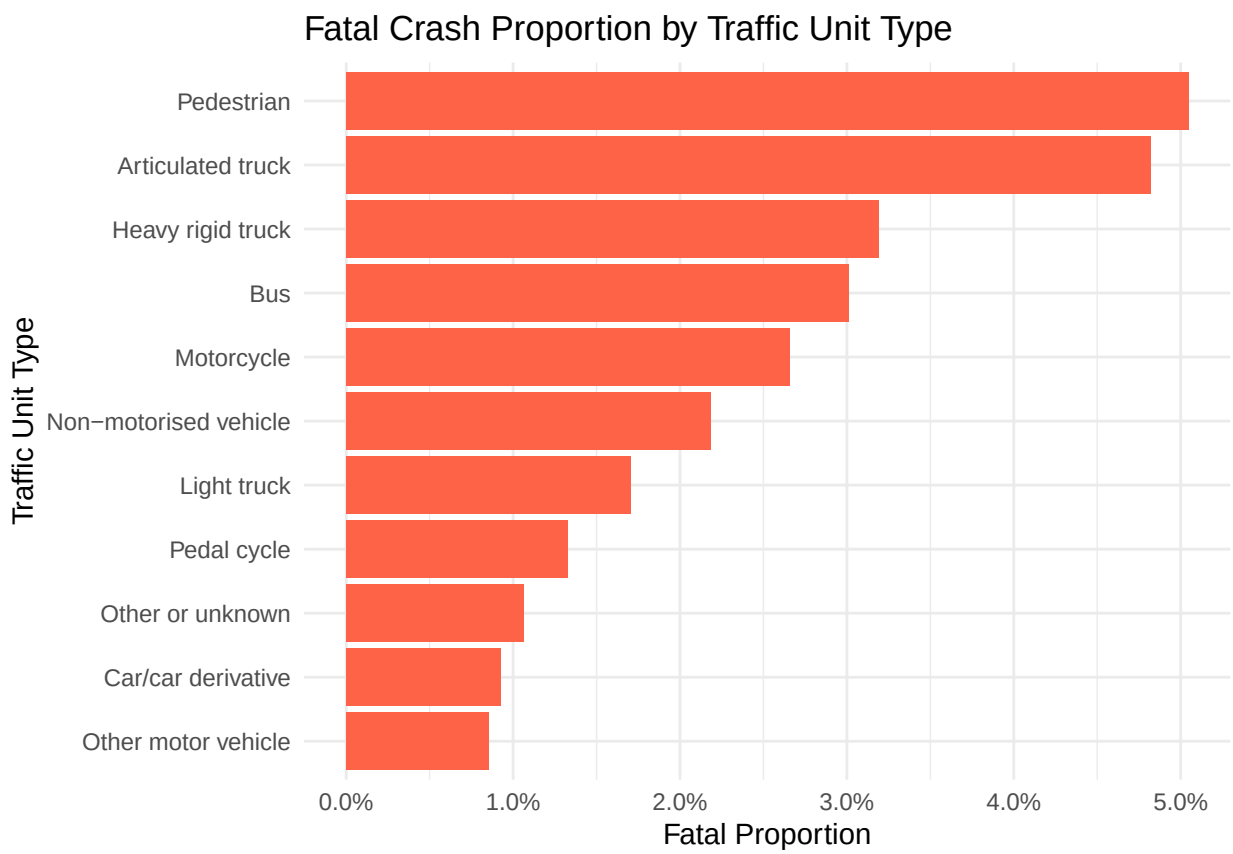


Crash counts are unevenly distributed across LGAs, indicating spatial heterogeneity in crash occurrence. This suggests that location-related variables such as LGA and urbanisation may add predictive value, although high-cardinality location variables may require grouping or regularisation during preprocessing.

3.3.6 Fatal-rate-tu-type

```
tu_fatal <- df_merged %>%
  filter(!is.na(`TU type group`), !is.na(`Degree of crash`)) %>%
  group_by(`TU type group`) %>%
  summarise(
    total = n(),
    fatal_rate = mean(`Degree of crash` == "Fatal"),
    .groups = "drop"
  ) %>%
  filter(total >= 100) %>%
  arrange(desc(fatal_rate))

ggplot(tu_fatal, aes(x = reorder(`TU type group`, fatal_rate), y = fatal_rate)) +
  geom_col(fill = "tomato") +
  coord_flip() +
  scale_y_continuous(labels = scales::percent_format(accuracy = 0.1)) +
  labs(
    title = "Fatal Crash Proportion by Traffic Unit Type",
    x = "Traffic Unit Type",
    y = "Fatal Proportion"
```

```
  ) +
  theme_minimal()
```



Fatal crash proportions vary noticeably across traffic unit types, suggesting that not all road users face the same severity risk. This reinforces the importance of traffic-unit-level features in the integrated dataset and supports the decision to retain traffic-unit characteristics for modelling.

4 Project Planning

4.1 Preprocessing Plan for handling your dataset's specific requirements to make it clean and ready for modelling.

To transform the raw NSW Road Crash Data into a clean, model-ready dataset, we propose the following preprocessing pipeline. All data-dependent steps (imputation, encoding, scaling, resampling) will be performed **within cross-validation folds** to prevent data leakage (James et al., 2021).

Step 1: Data Integration

The dataset consists of two linked tables — **CRASH** (92,189 rows × 50 columns) and **TRAFFIC UNIT** (170,962 rows × 10 columns) — connected via `Crash ID`. Since this is a one-to-many relationship, we will **aggregate** the TRAFFIC UNIT table to crash level before joining:

- Binary flags for vehicle type presence: `has_heavy_vehicle`, `has_motorcycle`, `has_pedestrian`, `has_bicycle`

- Count of distinct TU types per crash
- Binary flag for whether any object was hit

The aggregated table will be **left-joined** to CRASH on `Crash ID`, producing a single analysis-ready table.

Step 2: Feature Selection & Removal

Category	Columns	Reason
Identifiers	`Crash ID`	Not predictive
Target leakage	`Degree of crash - detailed`, `No. killed`, `No. seriously injured`, `No. moderately injured`, `No. minor-other injured`	Direct consequences of severity — would leak the target
Redundant location	`Street of crash`, `Identifying feature`, `Town`, `Route no.`, `Street type`, `Direction`	High cardinality; location already captured by `LGA`, `Lat/Lon`, `Urbanisation`
Near-zero variance	`Primary temporary feature` (98.5% NA), `Primary hazardous feature` (98.0% NA), `DCA supplement` (86.8% NA)	Too sparse to be informative

Retained features cover **environmental conditions** (weather, lighting, road surface, speed limit), **road characteristics** (alignment, type of location), **temporal patterns** (month, day of week, time interval), **spatial context** (urbanisation, LGA), and **traffic unit characteristics** (aggregated from TU table).

Step 3: Missing Value Handling

Variable	Missing %	Strategy
`Primary permanent feature`	73.0%	Convert to binary indicator `has_permanent_feature`; drop original
`Other TU type`	30.4%	Recode `NA` → `"Single vehicle"` (missing implies single-vehicle crash)
Remaining categorical	<1%	Treat `NA` as an explicit `"Unknown"` level
Remaining numeric	<1%	Median imputation (within training fold)

For categorical variables, treating missing as its own category is justified when the missingness pattern itself is informative (Little, 1988). We will verify this using `naniar::mcar_test()` and visualise with `visdat::vis_miss()`.

Step 4: Feature Engineering

Derived features to enhance predictive power:

- **Temporal**: `is_weekend` (Sat/Sun), `is_rush_hour` (7–9am / 4–6pm)
- **Environmental**: `is_night` (Darkness/Twilight), `is_wet` (non-Dry surface)
- **Speed bands**: Group `Speed limit` into `50`, `60-80`, `90-100`, `110+`, `Other`
- **Multi-vehicle**: Binary flag from `No. of traffic units involved > 1`

Step 5: Outlier Detection

- **Latitude/Longitude**: Remove records outside NSW bounds (lat: -37.5 to -28.0, lon: 140.0 to 154.0)
- **Distance**: Inspect for extreme values; winsorise at the 99th percentile if needed

Step 6: Encoding Categorical Variables

Encoding strategy is model-dependent:

- **Tree-based models** (Random Forest, XGBoost): Use R `factor` type directly — no encoding needed
- **Distance/linear models** (kNN, SVM, Logistic Regression, LDA): One-hot encoding for nominal variables; ordinal encoding for ordered variables (e.g., speed bands)
- **High-cardinality**: Pool `LGA` levels with <1% of observations into `"Other"` to reduce dimensionality

Step 7: Feature Scaling

- **kNN and SVM** are distance-based and sensitive to feature scale → apply **z-score standardisation** to all numeric features within each training fold
- **Tree-based models** are scale-invariant → no scaling needed

Step 8: Class Imbalance Handling

The target `Degree of crash` is highly imbalanced (Fatal: 1.5%, Injury: 66.4%, Non-casualty: 32.1%). Resampling will be applied **within the training fold only** to prevent leakage (Chawla et al., 2002):

1. **SMOTE** (`themis::step_smote`): Synthesise minority-class samples, especially for Fatal
2. **Down-sampling**: Undersample the majority class as a simpler alternative
3. Both strategies will be compared during model tuning to select the best approach

Step 9: Train/Test Split & Cross-Validation

- **Holdout**: 80/20 stratified split by `Degree of crash`
- **Nested CV**: Outer 5-fold for generalisation error; inner 5-fold for hyperparameter tuning
- All preprocessing steps will be embedded in a `tidymodels::recipe` pipeline so that transformations are fitted on training data only within each fold

Pipeline Summary

```
Raw Data (CRASH + TRAFFIC UNIT)

    1. Aggregate TRAFFIC UNIT → crash-level features
    2. Left join on Crash ID
    3. Remove identifiers, leakage vars, near-zero-variance cols
```

```
4. Feature engineering (binary flags, speed bands, temporal)
5. Outlier removal (geospatial bounds)
6. Train/Test split (80/20, stratified)

Within each CV fold (training only):
     7. Missing value handling
     8. Categorical encoding (one-hot / factor)
     9. Feature scaling (for kNN/SVM/LR)
     10. SMOTE / down-sampling for class imbalance
```

4.2 Model Plan: based on EDA, propose at least 5 classification models* suited to the data/problem with justification.

Based on the findings from EDA, we propose six classification models to predict **Degree of Crash** (a three-class outcome: Fatal, Injury, and Non-casualty). The EDA revealed that the target variable is extremely imbalanced—Fatal crashes account for a small proportion of the 92,189 observations—and that all predictor variables are categorical (e.g., `Weather`, `Natural lighting`, `Surface condition`, `Speed limit`, `Urbanisation`, `TU type group`). These characteristics directly inform the choice of models below.

All models will be implemented in R using the `caret` package for a unified training and cross-validation interface.

1. Multinomial Logistic Regression (Baseline)

- **Justification:** Proposed as our interpretable baseline model. It produces interpretable coefficients (log-odds) for each predictor category, which are meaningful for road safety stakeholders (e.g., quantifying how much driving at night increases the log-odds of a fatal crash relative to daylight). All subsequent models will be benchmarked against this baseline.
- **Implementation:** Categorical predictors will be encoded as dummy variables. We will use `multinom()` from the `nnet` package, utilising the `class_weight` argument to account for class imbalance.

2. Decision Tree Classifier (CART)

- **Justification:** Chosen for its full interpretability and ability to capture non-linear interactions between categorical features (e.g., capturing the joint rule "high speed limit AND rural area → higher fatal risk" in a single branch).
- **Implementation:** Implemented using the `rpart` package. Tree depth will be tuned via cross-validation to prevent overfitting, and the resulting tree will be visualised using `rpart.plot` to communicate high-risk conditions to non-technical audiences.

3. Random Forest Classifier

- **Justification:** Extends the decision tree by training an ensemble of trees on bootstrap samples, significantly reducing variance. It provides variable importance scores to identify which predictors most strongly impact crash severity.
- **Implementation:** With >90k observations, it is computationally feasible using the `randomForest` package. Class imbalance will be handled via the `classwt` argument in `randomForest()`.

4. Gradient Boosting Machine (XGBoost)

- **Justification:** Proposed as our primary high-performance model. By sequentially correcting misclassifications from prior trees, it is particularly effective at learning from the rare minority class (Fatal crashes) by exploiting subtle, high-value signals in the features.
- **Implementation:** Implemented using the `xgboost` package. Hyperparameters (`nrounds`, `max_depth`, `eta`) will be tuned via `caret::trainControl`. The `scale_pos_weight` parameter will be adjusted to address the class imbalance.

5. K-Nearest Neighbours (KNN)

- **Justification:** Included as a non-parametric, instance-based comparison model to test whether crash severity is locally consistent in feature space (i.e., whether crashes occurring under similar conditions tend to share the same outcome).
- **Implementation:** Implemented using the `class` package. Since all features are categorical, the Gower distance metric (via the `gower` package) will be used. The optimal value of k will be selected via cross-validation.

6. Naive Bayes

- **Justification:** A natural fit for datasets where all features are categorical. It serves as a strong probabilistic benchmark: if more complex models cannot substantially outperform it, this suggests that the individual distributions of features (e.g., Weather, Speed limit) already carry most of the predictive information.
- **Implementation:** Implemented using the `e1071` package. Under the conditional independence assumption, it is extremely fast to train and provides probability estimates for each severity class.

4.3 Evaluation Plan: propose and justify performance metrics for final model

evaluation.

Standard accuracy is an inadequate evaluation metric for this problem due to severe class imbalance. A classifier predicting *Injury* for every observation would achieve high accuracy while completely failing to identify any *Fatal* crash. The metrics below are selected to reflect both overall classifier performance and the disproportionate importance of correctly identifying fatal outcomes.

Performance Metrics

- **Macro-Averaged F1 Score (Primary):** Computes the F1 score for each of the three classes independently and then averages them with equal weight. This ensures that performance on the rare *Fatal* class receives the same weight as the dominant *Injury* class. This will be our primary metric for ranking models.
- **Fatal Class Recall (Primary Constraint):** Measures the proportion of actual *Fatal* crashes correctly identified. A model with high Macro F1 but low Fatal recall will be deprioritised, as failing to flag potentially fatal conditions is the most consequential error.
- **Weighted F1 Score (Secondary):** Averages F1 per class weighted by each class's support. It provides a realistic picture of overall performance on the true class distribution.
- **Cohen's Kappa (κ) (Secondary):** Adjusts accuracy for the agreement expected by chance. Computed directly by `caret::confusionMatrix()`, providing a clear sanity check (values near 0 indicate performance no better than random guessing).

- **Confusion Matrix:** Evaluated via `caret::confusionMatrix()` and visualised as a heatmap using `ggplot2` to explicitly monitor misclassification directions (e.g., dangerous errors where Fatal crashes are predicted as Injury).

Validation Strategy

- **Cross-Validation:** Stratified 5-fold cross-validation will be applied to the training data using `caret::trainControl(method = "cv", number = 5, classProbs = TRUE, savePredictions = TRUE)`. This ensures class proportions are preserved across all folds.
- **Temporal Split:** We will use a time-based holdout strategy: **2020–2023 data for training/CV** and **2024 data as the held-out test set**. This reflects real-world deployment conditions where historical data predicts future crashes.
- **Unbiased Estimation:** The final held-out test set (2024) will be evaluated *once only* after model selection and tuning are complete to obtain an unbiased estimate of generalisation performance.

4.4 Timeline and Project Milestones

The project timeline has been structured in alignment with the assignment deadlines to ensure clear and systematic progression of tasks from understanding the dataset up-to final model development and reporting.

1-7 April: Problem Framing and Data Acquisition

Different datasets were evaluated and a final dataset was selected, along with finalization of the research question. The crash and Traffic Unit datasets (2020 -2024) were sourced from the NSW Government open data platform. Initial inspection was carried out to understand the dataset structure, key variables, and differences.

7-10 April: Data Integration and Initial Data Preparation

The crash and traffic unit datasets were merged using Crash ID, resulting in a unified dataset containing 170,962 records at the vehicle level. Initial data preparation involved examining missing values, identifying inconsistencies introduced via the merging, and understanding different variable types. Full preprocessing and feature engineering will be conducted in the modelling phase.

10-14 April: Exploratory Data Analysis (EDA)

Univariate and bivariate analyses were performed to explore the distribution and relationship between variables. This includes examining environmental factors (weather, lighting), temporal variables(time of day, day of week), and vehicle- level characteristics (vehicle type, manoeuvre). Insights from EDA were used to inform feature selection and guide model planning.

14-17 April: Feature Planning and Model Design

Based on EDA findings, relevant features were identified and transformations were planned, including the plan to encode categorical variables and aggregate vehicle-level information. A set of classification models were also selected for implementation in the next phase.

17-19 April: Plan & EDA Report Finalization

All findings were compiled into a structured R Markdown report and submitted by the deadline.

6–10 May: Data Preprocessing and Baseline Modelling

Preprocessing steps such as handling missing values, encoding categorical variables, and feature engineering will be carried out. The baseline model (Multinomial Logistic Regression) will also be implemented and evaluated.

11–15 May: Advanced Model Development and Tuning

Advanced models will be implemented and refined, including hyperparameter tuning via cross-validation to improve performance.

16–20 May: Model Evaluation and Report Writing

Models will be evaluated using appropriate performance metrics, considering class imbalance in crash severity. The final model will be selected and justified. Results, insights, and interpretations will be consolidated into the final report.

21–24 May: Report Refinement

Final refinements will be made to ensure clarity, consistency, and adherence to the submission requirements.

25–31 May: Final Checks and Contribution Review

Team contributions will be reviewed and documented. A final check of the report will be conducted before submission.

Backup Plan

The team acknowledges that unforeseen circumstances such as illness, personal emergencies, or other commitments may arise during the project period. With this in mind, the following contingency measures have been put in place. The workload of members has been structured so that 2 members are on standby at any given time, ready to take over tasks if a team member is unable to meet a deadline. The remaining members will provide 2 check-ins per week to monitor task progress and flag any delays early. If a task is not completed by its deadline, it will be immediately handed over to a standby member to minimise disruption to the overall timeline. All work will be stored in a shared github repository accessible to all team members to ensure continuity.

The team will hold 2 meetings per week during project periods. One every Thursday in-person after the workshop, and one every Monday via Zoom to track progress, update team members, and provide support. Each member is expected to contribute approximately 20% to the overall project across both phases.

References

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Acknowledgement of AI

Group 07 (Workshop 12) declares that all ideas, analysis, and written content presented in this report are our own. We did not use generative AI to write our code. We used ChatGPT (GPT-5.3, OpenAI, https://chatgpt.com) and Claude Sonnet 4.5 (Anthropic, https://claude.ai) for code refinement, grammar checking, and reviewing our report against the marking rubric.